

Optical & Magneto-Optical Signatures of Altermagnetism

PROBING THE EFFECT OF SPIN-SPLITTING

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Hauptseminar: Talk 8 of 12

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University of Stuttgart
Germany

How do we identify Altermagnets?

The Identification Challenge

Initial Claims (Surface)

- ARPES observed apparent spin-splitting in RuO_2 .
- Led to widespread classification as a model altermagnet.

The Debunking (Bulk)

- Optical spectroscopy revealed **Fermi-liquid behavior**.
- Matches a non-magnetic model better than an altermagnetic one.

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Altermagnetism in RuO_2 thin films: A new magnetic material for the AI era

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Article | [Open access](#) | Published: 24 September 2025

Evidence for single variant in altermagnetic $RuO_2(101)$ thin films

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VS



Fermi-liquid behavior of nonaltermagnetic RuO_2

Maxim Wenzel,¹ Ece Uykur^{2,*}, Sahana Rößler³, Marcus Schmidt,³ Oleg Janson⁴, Achyut Tiwari,¹ Martin Dressel¹ and Alexander A. Tsirlin^{5,†}

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³*Max Planck Institute for Chemical Physics of Solids, D-01067 Dresden, Germany*

⁴*Institute for Theoretical Solid State Physics, Leibniz IFW Dresden, D-01069 Dresden, Germany*

⁵*Felix Bloch Institute for Solid-State Physics, University of Leipzig, D-04103 Leipzig, Germany*

Revisiting altermagnetism in RuO_2 : a study of laser-pulse induced charge dynamics by time-domain terahertz spectroscopy

 Check for updates

David T. Plouff[†] , Laura Scheuer[†], Shreya Shrestha[†], Weipeng Wu[†], Nawsher J. Parvez[†], Subhash Bhatt[†], Xinhao Wang[†], Lars Gundlach^{†,‡} , M. Benjamin Jungfleisch[†]  & John Q. Xiao[†] 

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Ambiguity remains: Surface effect or Bulk property?
We need a **definitive Bulk Fingerprint**.

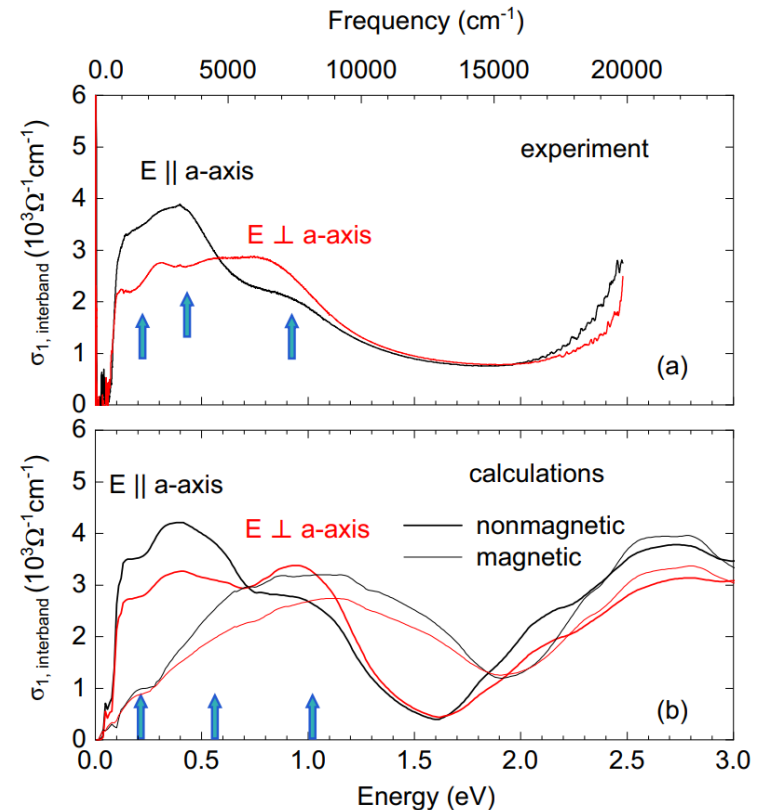
Fermi-liquid behavior of nonaltermagnetic

NON-MAGNETIC SIMULATION MATCHES THE EXPERIMENTAL DATA RuO2

DFT: Magnetic/Non-Magnetic vs Experiment

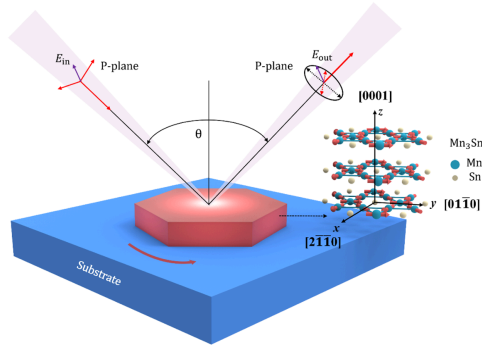
"Adding magnetism clearly deteriorates the description of the experimental optical conductivity."

Note in particular the acute discrepancies at low energies where the sharp increase in the optical conductivity (Pauli edge) cannot be reproduced by the magnetic solution."

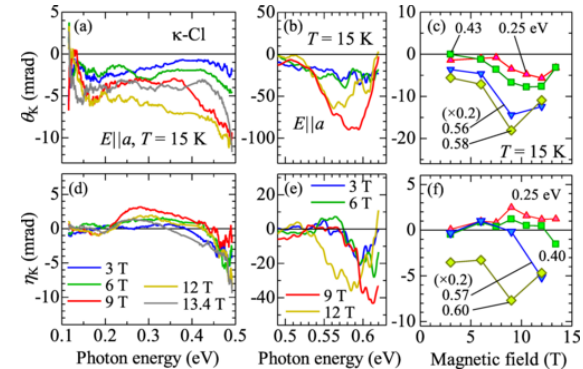
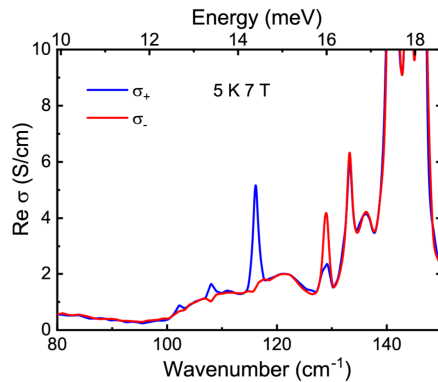


ARPES predicted Altermagnetism
Magneto-Optical measurements show otherwise!

Background: Magneto Optical Kerr Effect & Mueller Matrix



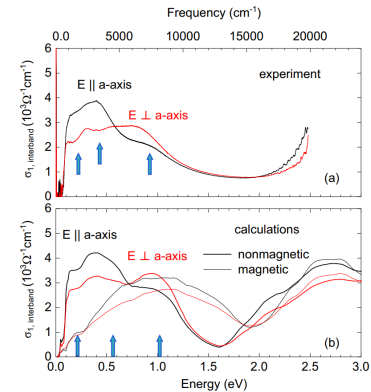
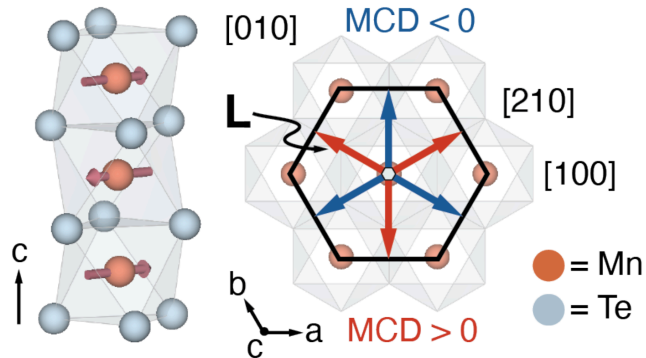
$$S_{out}(\omega) = \begin{pmatrix} M_{11} & M_{12} & M_{13} \\ M_{21} & M_{22} & M_{23} \\ M_{31} & M_{32} & M_{33} \\ M_{41} & M_{42} & M_{43} \end{pmatrix}$$



Case Studies: Off-Diagonal Elements of

$$\tilde{\sigma}_{ij}$$

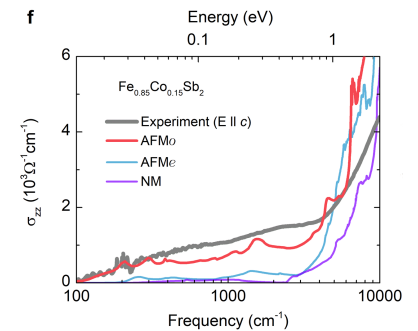
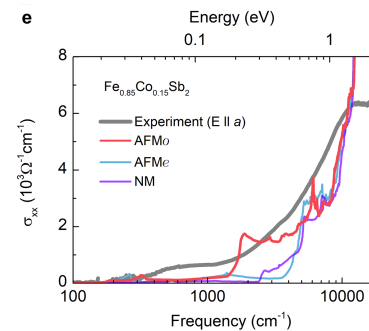
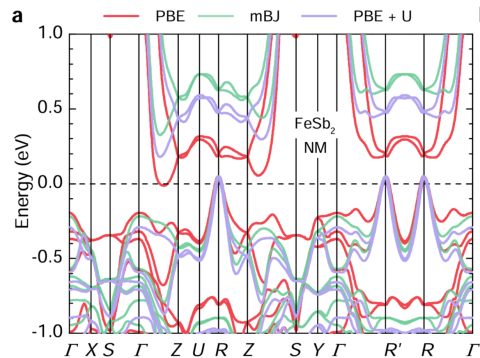
$$\tilde{\sigma} = \begin{pmatrix} \sigma_{xx} & \sigma_{xy} & 0 \\ -\sigma_{xy} & \sigma_{xx} & 0 \\ 0 & 0 & \sigma_{zz} \end{pmatrix}$$



Case Studies: Diagonal Elements of

$\tilde{\sigma}_{ij}$

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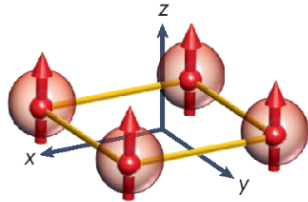


Breaking the Symmetry

TRANSFORMATION SYMMETRY IN DIFFERENT MAGNETIC ORDERS

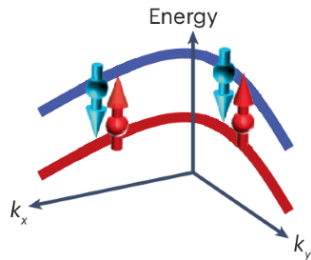
A: FERRO

- $\mathbf{M} \neq 0$
- **Breaks** time-reversal symmetry $\mathcal{T}$
- No spin degeneracy



Momentum space

Unidirectional splitting

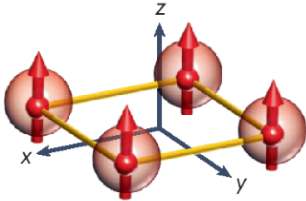


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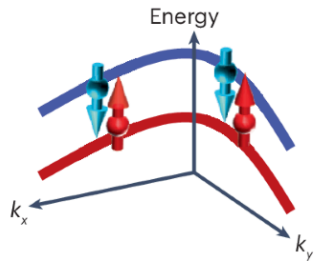
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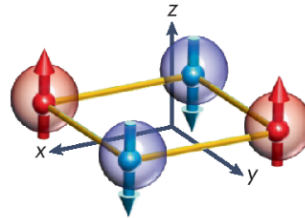
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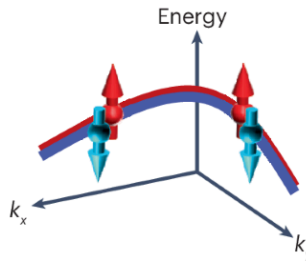


B: ANTI-FERRO

- $M = 0$
- **Preserves** symmetry PT
- Spin-degenerate electronic bands



Non-splitting

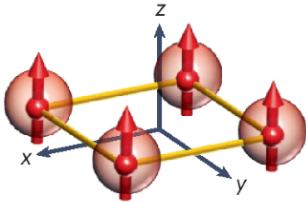


Breaking the Symmetry

TRANSFORMATION SYMMETRY IN DIFFERENT MAGNETIC ORDERS

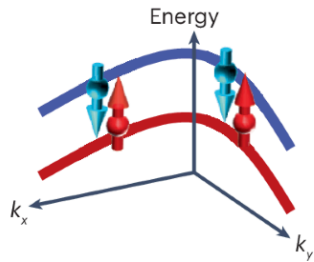
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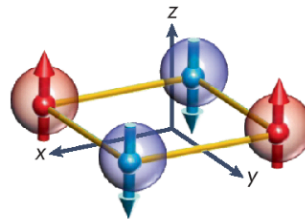
Momentum space

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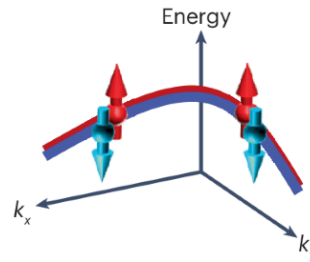


B: ANTI-FERRO

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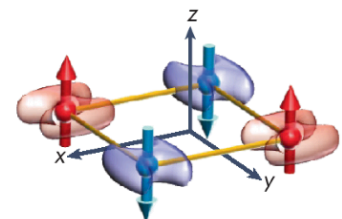


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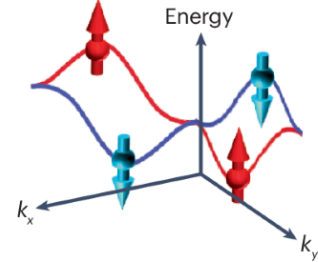


C: ALTERMAGNET

- $M = 0$
- **Breaks** $\mathcal{T}$ but **preserves** $\mathcal{PT}$
- No spin degeneracy



Alternating splitting



Optical Behavior

LINEAR RESPONSE THEOREM AND THE OPTICAL CONDUCTIVITY TENSOR

Normal materials follow **Linear Response Theorem:**

$\tilde{\sigma}_{ij}(\omega)$: complex optical conductivity tensor
 $\tilde{\epsilon} \leftrightarrow \tilde{\sigma}$

$$\begin{cases} \tilde{\epsilon}_{xx} = 1 + \frac{i\tilde{\sigma}_{xx}}{\epsilon_0\omega} \\ \tilde{\epsilon}_{xy} = \frac{i\tilde{\sigma}_{xy}}{\epsilon_0\omega} \end{cases}$$

ORDINARY:

MAGNETIC:

$$\tilde{\sigma} = \begin{pmatrix} \sigma_{xx} & 0 & 0 \\ 0 & \sigma_{xx} & 0 \\ 0 & 0 & \sigma_{xx} \end{pmatrix}$$

STRUCTURAL:

$$\tilde{\sigma} = \begin{pmatrix} \tilde{\sigma}_{xx} & \tilde{\sigma}_{xy} & 0 \\ -\tilde{\sigma}_{xy} & \tilde{\sigma}_{xx} & 0 \\ 0 & 0 & \tilde{\sigma}_{zz} \end{pmatrix}$$

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$\tilde{\sigma}_{ij}(\omega)$: complex optical conductivity tensor

- **Absolute Reflectivity** $R(\omega)$

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- **Rotation** $\Delta\Psi(\omega)$, **Ellipticity** η

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⚠Problem:

Isolate Altermagnetism as source of off-diagonal elements

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Optical Behavior

MECHANISMS GENERATING OFF-DIAGONAL TENSOR ELEMENTS ()

Antisymmetric ()

Requires Time-Reversal Breaking $\sigma_{xy} = -\sigma_{yx}$

- **External Magnetic Field ():** Explicitly breaks $\mathcal{T}$ via the Lorentz force, cyclotron motion / $\mathbf{B} \neq 0$. Ordinary Hall effect.
- **Ferromagnetism ():** Spontaneous global -breaking coupled with Spin-Orbit Interaction $\mathbf{M} \neq 0$.
- **Altermagnetism ():** Spontaneous -breaking through alternating crystal symmetries. $\mathbf{M} = 0$.

Symmetric ()

Structural & Reciprocal Origins $\sigma_{xy} = \sigma_{yx}$

- **Crystal Misalignment:** Choosing a laboratory coordinate framework that is rotated relative to the principal crystallographic axes.
- **Linear Birefringence / Dichroism:** Intrinsic optical anisotropy in lower-symmetry crystal systems (like the orthorhombic structure of FeSb_2).
- **Piezomagnetic / Magnetoelectric Effects:** Strain-induced or voltage-induced modifications to the tensor framework.

Optical Behavior

MECHANISMS GENERATING OFF-DIAGONAL TENSOR ELEMENTS (Σ_{XY})

Antisymmetric ($\sigma_{xy} = -\sigma_{yx}$)

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Optical Behavior

STRUCTURAL ORIGIN OF : LOW-SYMMETRY MONOCLINIC/ORTHORHOMBIC SYSTEMS

Example: Structural Monoclinic Angle Alignment

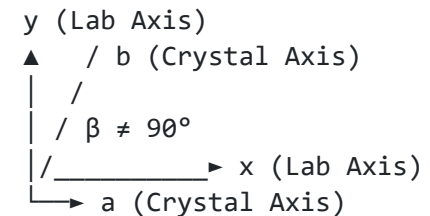
Internal crystal axes do not match an orthogonal laboratory coordinate framework

(a, b, c)
 (x, y, z)

- **Purely Structural:** Occurs in completely non-magnetic, time-reversal symmetric insulators.
- **The Consequence:** Electric fields applied along the laboratory x -axis naturally induce a polarization current along the laboratory y -axis due to the tilted unit cell shape.
- **Tensor Property:** Enforces a strictly **symmetric** off-diagonal element: .

$$\sigma_{xy} = \sigma_{yx}$$

Laboratory vs. Crystal Coordinates



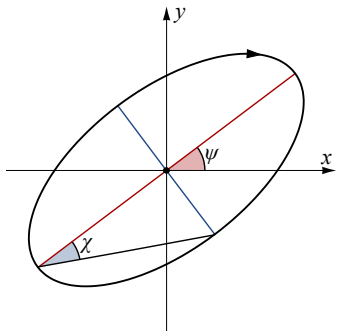
Applying E_x yields a projection along both a and b , mechanically generating an response J_y .

The Magneto-Optical Kerr Effect (MOKE)

MEASURING THE OPTICAL CONDUCTIVITY TENSOR

$$\tilde{\sigma}_{xy}(\omega) \propto \theta_K + i\eta_K$$

- (Rotation) Dispersive part, $\text{Im}()$
 $\theta_K(\Psi) \leftrightarrow \tilde{\sigma}_{xy}$
- (Ellipticity) Dissipative part, $\text{Re}()$
 $\eta_K \leftrightarrow \tilde{\sigma}_{xy}$

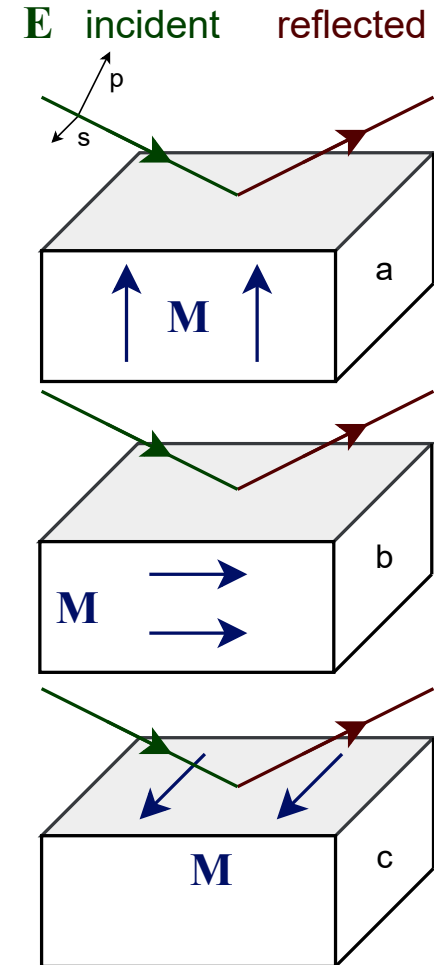


CONVENTIONS:

- **a: polar**
- **b: longitudinal**
- **c: transversal**

Polar MOKE: light interacts only with the xy -plane. Lets us extract from the reflected θ_K and η_K .

$$\sigma_{xy} \propto \theta_K + i\eta_K$$



The Magneto-Optical Kerr Effect (MOKE)

MEASURING THE KERR ROTATION AND ELLIPTICITY

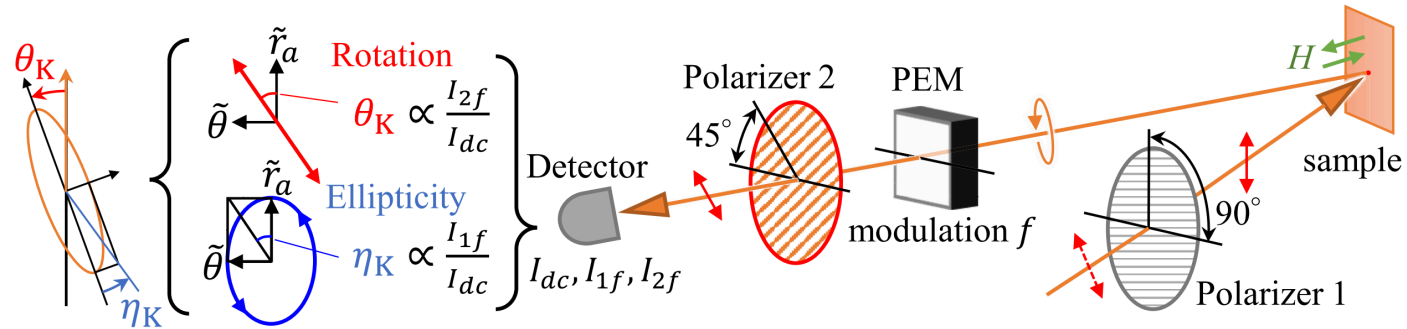
The Magneto-Optical Kerr Effect (MOKE)

MEASURING THE KERR ROTATION AND ELLIPTICITY

INSERT DUMMY PLOT OF ANY MEASUREMENT

Measuring the MOKE Response

EXPERIMENTAL SETUP AND DATA ANALYSIS: SINGLE FREQUENCY



SETUP:

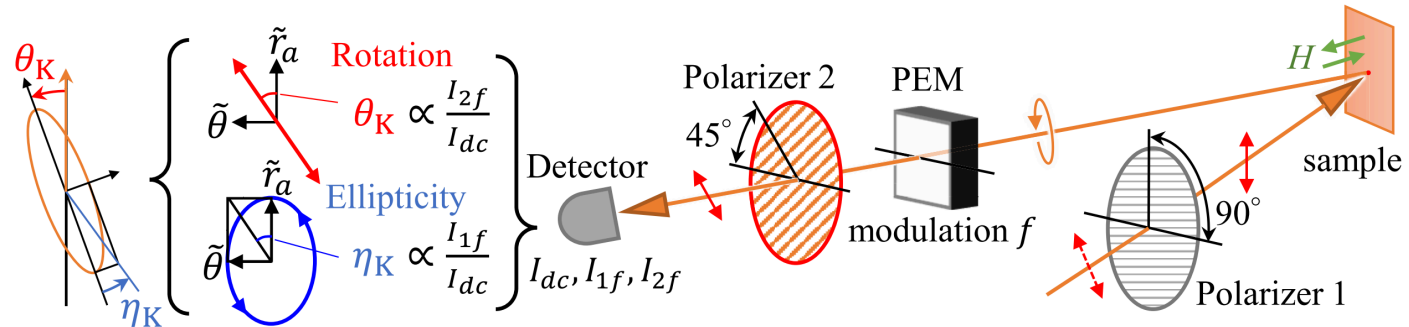
- **Light source:** Monochromatic, linearly polarized light
- **PEM:** Polarization-encoding module
- **Analyzer:** Rotating polarizer
- **Detector:** Intensity measurement device
- **Lock-in Amplifier:** Optional, Signal processing unit

DATA ANALYSIS:

- I_{dc} : Direct current, Absolute Reflectivity $R(\omega)$
- I_{1f} : First harmonic component, Kerr ellipticity η_K
- I_{2f} : Second harmonic component, Kerr rotation θ_K

Measuring the MOKE Response

EXPERIMENTAL SETUP AND DATA ANALYSIS: SINGLE FREQUENCY



ADVANTAGES:

- **Sensitivity:** High, up to milliradians
- **Noise:** Low, because of Lock-In Amplifier

DISADVANTAGES:

- **Chromatic hardware:** Wavelength dependent (e.g. PEM)
- **Resolution:** Low, maybe one frequency

Useless or very high effort for broad band measuring!

Optical Behavior

LINEAR RESPONSE THEOREM

CONCEPT

For broadband spectrum, **remove all wavelength-dependent hardware** (like waveplates and PEMs), use wideband linear polarizers.

PROBLEM

Linear polarizers can measure absolute reflectivity and Kerr rotation but without a waveplate, R **we cannot measure** θ_K **ellipticity!**

Optical Behavior

LINEAR RESPONSE THEOREM

CONCEPT

Remove all wavelength-dependent
hardware

PROBLEM

We cannot measure ellipticity!

Optical Behavior

LINEAR RESPONSE THEOREM

CONCEPT

Remove all wavelength-dependent hardware

PROBLEM

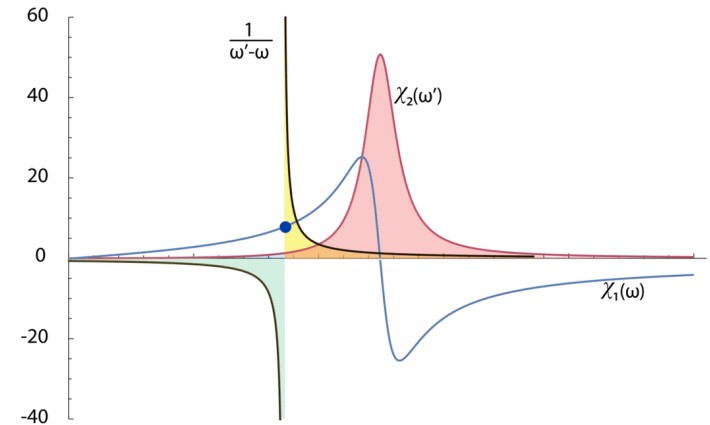
We cannot measure ellipticity!

KRAMERS KRONIG RELATION

Condition: Response function (analytic in the upper half-plane).

$$\text{Re}[f(\omega)] = \frac{1}{\pi} \mathcal{P} \int_{-\infty}^{\infty} \frac{\text{Im}[f(x)]}{x - \omega} dx$$

$$\text{Im}[f(\omega)] = -\frac{1}{\pi} \mathcal{P} \int_{-\infty}^{\infty} \frac{\text{Re}[f(x)]}{x - \omega} dx$$



<https://upload.wikimedia.org/wikipedia/commons/4/45/Kramersgifwiki.gif>: (2026)

Magneto-Optical Kramers-Kronig Analysis

EXPERIMENTAL SETUP AND DATA ANALYSIS: BROAD BAND

- **Measurement:**

Use linear polarizers to get total reflectivity and Kerr rotation .

$$R(\omega) \quad \theta_K(\omega)$$

- **Define Auxiliary Function:**

- **Apply Kramers-Kronig:**

$$f(\omega) = \ln \left(\frac{\tilde{r}_-}{\tilde{r}_+} \right) = \ln c(\omega) + i \cdot 2\theta_K(\omega)$$

$$\ln c(\omega) = \frac{2\omega}{\pi} \mathcal{P} \int_0^\infty \frac{2\theta_K(x)}{x^2 - \omega^2} dx$$

- **Reconstruct & Solve:**

With $\tilde{r}_+$ and $\tilde{r}_-$, we solve for R_+ and R_- individually. A final KK transform on these yields the full complex .

$$[R_+ = \frac{1}{2}(R_+ + R_-) \quad R_- = \frac{1}{2}(R_+ - R_-)] \quad \tilde{r}_\pm(\omega)$$

Magneto-Optical Kramers-Kronig Analysis

EXPERIMENTAL SETUP AND DATA ANALYSIS: BROAD BAND

- **Auxiliary Function:**

- **Kramers-Kronig:**
$$f(\omega) = \ln \left(\frac{\tilde{r}_-}{\tilde{r}_+} \right) = \ln c(\omega) + i \cdot 2\theta_K(\omega)$$

- **Reconstruct & Solve:**
$$\ln c(\omega) = \frac{2\omega}{\pi} \mathcal{P} \int_0^\infty \frac{2\theta_K(x)}{x^2 - \omega^2} dx$$

But what if we measure in extreme environments?

$$R = \frac{R_+ + R_-}{2}$$

$$c = \frac{R_-}{R_+}$$

High magnetic fields & low temperatures obstruct rotating equipment!

ADVANTAGE

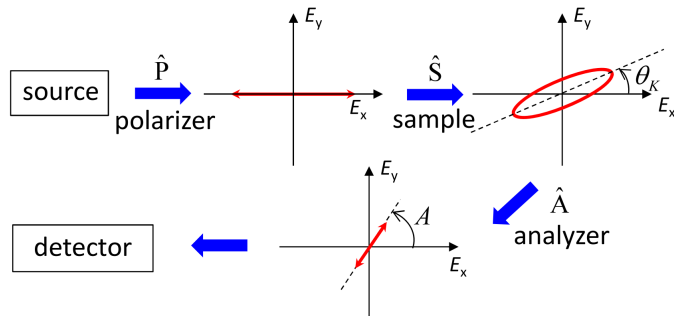
- Avoids the spectral limitations of wave plates
- Still highly sensitive to small signals
- Ideal for altermagnets

MOKKA: The 45-Degree Fast Protocol

BROAD BAND MEASURING IN EXTREME ENVIRONMENTS

Setup

- Polarizer and Analyzer are fixed at 45° to each other
- This geometry **maximizes sensitivity** to rotation



Analysis

- Detector intensity proportional to the Kerr angle, for small angles:

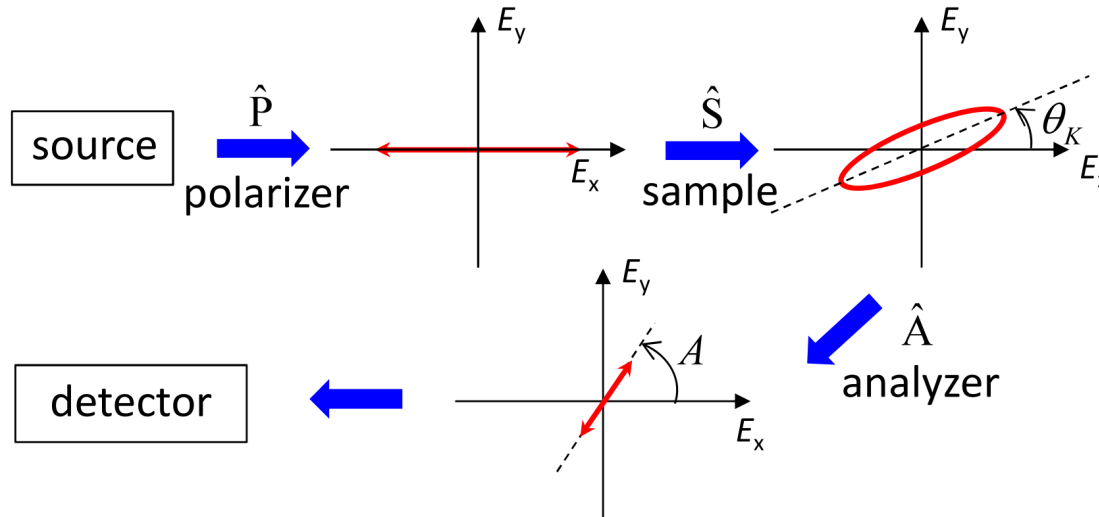
$$I_{\pm 45^\circ} \approx C \cdot R(\omega) \frac{1 \pm 2\theta_K}{2}$$

- Measure intensity ratio at positive and negative magnetic fields and isolate:

$$\rho_{\pm 45^\circ}(B) = \frac{I_{\pm 45^\circ}(+B)}{I_{\pm 45^\circ}(-B)}$$

MOKKA: The 45-Degree Fast Protocol

BROAD BAND MEASURING IN EXTREME ENVIRONMENTS



Disadvantage

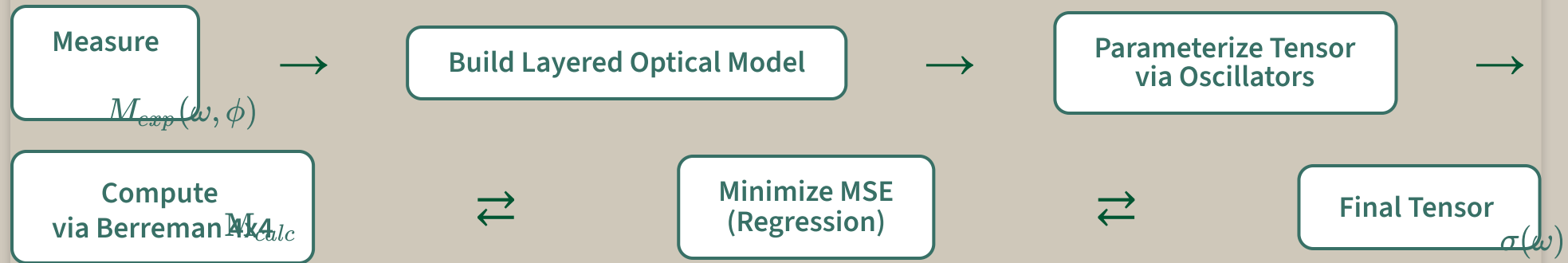
Limited spectral resolution and sensitivity compared to other techniques.

Advantage

Completely fit for comparing results with DFT calculations

Mueller Matrix Analysis

DECOMPOSING THE OPTICAL RESPONSE



Mueller Matrix Analysis

DECOMPOSING THE OPTICAL RESPONSE



Mueller Matrix Analysis

DECOMPOSING THE OPTICAL RESPONSE



Mueller Matrix Formula:

Measure the Mueller Matrix by varying the angle of incidence (Φ) and scanning the photon energy ($\hbar\omega$):

$$S_{\text{out}}(\omega) = M(\omega) \cdot S_{\text{in}}(\omega) = \begin{pmatrix} M_{11} & M_{12} & M_{13} & M_{14} \\ M_{21} & M_{22} & M_{23} & M_{24} \\ M_{31} & M_{32} & M_{33} & M_{34} \\ M_{41} & M_{42} & M_{43} & M_{44} \end{pmatrix} \begin{pmatrix} I \\ Q \\ U \\ V \end{pmatrix}$$

Mueller Matrix Analysis

DECOMPOSING THE OPTICAL RESPONSE



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- ordinary refractive index and thickness data.

Mueller Matrix Analysis

DECOMPOSING THE OPTICAL RESPONSE



Mueller Matrix Formula:

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- ordinary refractive index and thickness data.
- cross-polarization induced by the off-diagonal elements of your optical conductivity tensor

Mueller Matrix Analysis

DECOMPOSING THE OPTICAL RESPONSE



- ordinary refractive index and thickness data.

$$S_{out}(\omega) = \begin{pmatrix} M_{11} & M_{12} & M_{13} & M_{14} \\ M_{21} & M_{22} & M_{23} & M_{24} \\ M_{31} & M_{32} & M_{33} & M_{34} \\ M_{41} & M_{42} & M_{43} & M_{44} \end{pmatrix} \bullet \begin{pmatrix} I \\ Q \\ U \\ V \end{pmatrix}$$

● cross-polarization induced by the off-diagonal elements of your optical conductivity tensor

Mueller Matrix Analysis

DECOMPOSING THE OPTICAL RESPONSE



$$S_{\text{out}}(\omega) = \begin{pmatrix} M_{11} & M_{12} & M_{13} & M_{14} \\ M_{21} & M_{22} & M_{23} & M_{24} \\ M_{31} & M_{32} & M_{33} & M_{34} \\ M_{41} & M_{42} & M_{43} & M_{44} \end{pmatrix} \begin{pmatrix} I \\ Q \\ U \\ V \end{pmatrix}$$

- ordinary refractive index and thickness data.
- cross-polarization induced by the off-diagonal elements of your optical conductivity tensor

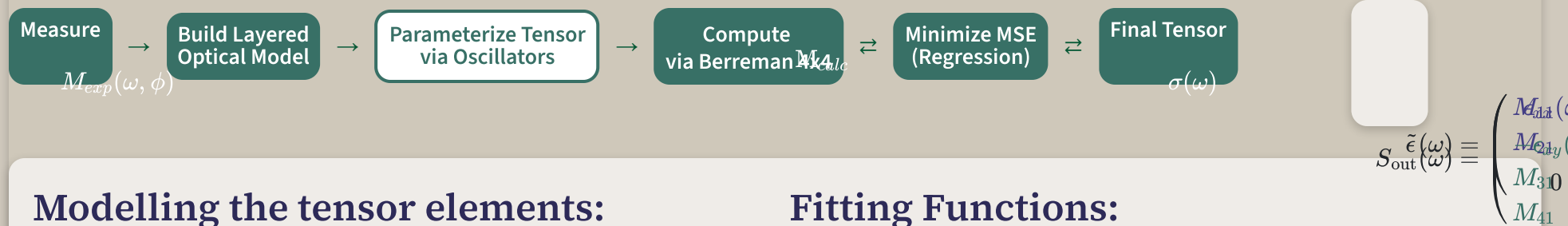
Geometric Model Construction of the Sample Stack

[Ambient (Air)] → [Surface Roughness] → [Anisotropic/MO Thin Film] → [Substrate]

Typical dielectric tensors for the material layers:

$$\tilde{\epsilon}(\omega) = \begin{pmatrix} \epsilon_{xx}(\omega) & \epsilon_{xy}(\omega) & 0 \\ -\epsilon_{xy}(\omega) & \epsilon_{yy}(\omega) & 0 \\ 0 & 0 & \epsilon_{zz}(\omega) \end{pmatrix}$$

Mueller Matrix Analysis



Modelling the tensor elements:

- Lorentz oscillators
- Tauc-Lorentz models (semiconductors/insulators)
- Drude terms (metals)

Fitting Functions:

$$\epsilon_{xy}(\omega) = \sum_j \left[\frac{A_j^+}{\omega_{0,j}^+ - \omega - i\Gamma_j^+} - \frac{A_j^-}{\omega_{0,j}^- - \omega - i\Gamma_j^-} \right]$$

$$\epsilon_{xx}(\omega) = \epsilon_{\infty}^j + \sum_k \frac{A_k^+}{\omega_{0,k}^2 - \omega^2 - i\Gamma_k \omega}$$

System is reduced to a finite set of global fit parameters:

amplitudes (A), **resonance energies** (ω_0), and **broadenings** (Γ)

Mueller Matrix Analysis



$$S_{\text{out}}(\omega) = \begin{pmatrix} M_{11}(\omega) & M_{12}(\omega) & M_{13}(\omega) & M_{14}(\omega) \\ M_{21}(\omega) & M_{22}(\omega) & M_{23}(\omega) & M_{24}(\omega) \\ M_{31}(\omega) & M_{32}(\omega) & M_{33}(\omega) & M_{34}(\omega) \\ M_{41}(\omega) & M_{42}(\omega) & M_{43}(\omega) & M_{44}(\omega) \end{pmatrix} \begin{pmatrix} I \\ Q \\ U \\ V \end{pmatrix}$$

$$\epsilon_{xx}(\omega) = \epsilon_{\infty} + \sum_j \frac{A_j}{\omega_{0,j}^2 - \omega^2 - i\Gamma_j \omega}$$

$$\epsilon_{xy}(\omega) = \sum_j \frac{B_j}{\omega_{0,j}^2 - \omega^2 - i\Gamma_j \omega}$$

Regression Analysis and Extracting the Conductivity Tensor

- Vary oscillator parameters and layer thicknesses to minimize a global Mean Squared Error (MSE)
- Extract the complex optical conductivity tensor from the dielectric tensor

$$\tilde{\sigma}(\omega) \quad \tilde{\epsilon}(\omega)$$

$$\tilde{\sigma}(\omega) = -i\epsilon_0\omega [\tilde{\epsilon}(\omega) - \mathbf{I}]$$

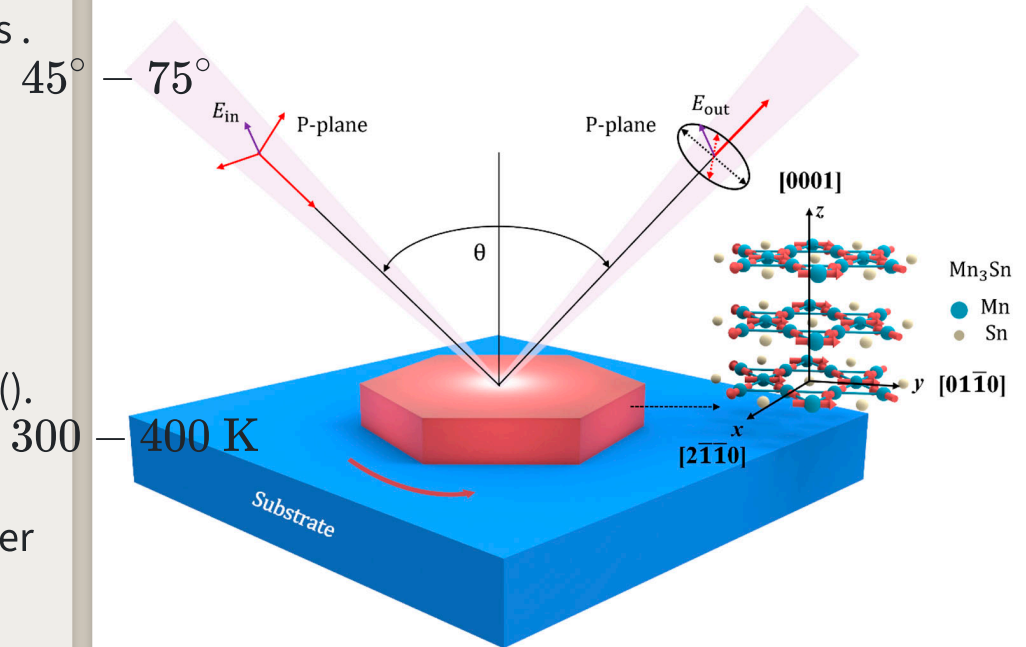
Mueller Matrix Measurement

ANTIFERROMAGNETIC WEYL SEMI-METAL



Setup

- **Optical Range:** Spectroscopic Ellipsometry spanning (UV to Near-IR) over incident angles .
300 — 1690 nm
- **Magnetic Control:** In-plane Voigt Geometry using a permanent magnet aligned along specific crystalline orientations.
- **Thermal Control:** A custom semiconductor stage enabling variable-temperature sweeps ().
300 — 400 K
- **Inversion:** Convert Mueller Matrix into anisotropic permittivity tensor via the Transfer Matrix Method.
 4×4

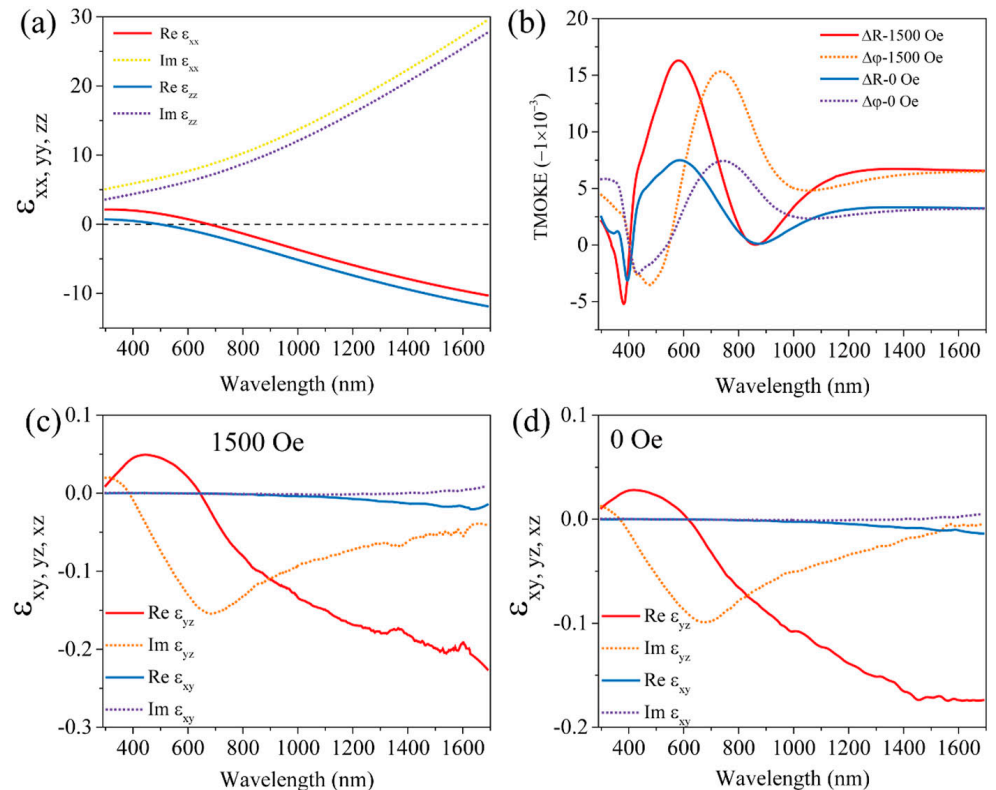


Mueller Matrix Measurement

DIRECT MAPPING OF CRYSTAL ANISOTROPY & BERRY CURVATURE

Analysis

- **Diagonal Response ($\epsilon_{xx}, \epsilon_{yy}, \epsilon_{zz}$):** Maps the fundamental, highly anisotropic electronic transitions along different crystal axes
- **Off-Diagonal Response ($\epsilon_{xy}, \epsilon_{yz}, \epsilon_{zx}$):** Captures the spontaneous magneto-optical activity
- **The Metric:** Evaluated through the Transverse Magneto-Optical Kerr Effect (TMOKE) response across the spectrum.

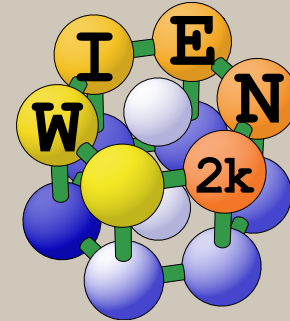


Simulation methods

QUANTUM MECHANICAL SIMULATION AS REFERENCE TO EXPERIMENTAL RESULTS

Optical Response from Experiments vs Predictions from DFT Simulations

- Calculate band structure
- Use Kubo formula to calculate optical conductivity tensor
- Compare with experimental MOKE data to confirm alternating magnetic order



We need a path from band structure to optical response:

Kubo Formula

The Kubo Formula

LINKING BAND STRUCTURE TO OPTICAL RESPONSE

$$\tilde{\sigma}_{xy}(\omega) \propto \frac{1}{\omega} \sum_{n,m} \int_{\text{BZ}} dk \frac{f(E_{n,\{k\}}) - f(E_{m,k})}{E_{m,k} - E_{n,k} - \hbar\omega - i\Gamma} \times \langle \psi_n |$$

The Kubo Formula

LINKING BAND STRUCTURE TO OPTICAL RESPONSE

$$\tilde{\sigma}_{xy}(\omega) \propto \frac{1}{\omega} \sum_{n,m} \int_{\text{BZ}} d\mathbf{k} \frac{f(E_{n,\{\mathbf{k}\}}) - f(E_{m,\mathbf{k}})}{E_{m,\mathbf{k}} - E_{n,\mathbf{k}} - \hbar\omega - i\Gamma} \times \langle \psi_{n,\mathbf{k}} | \hat{v}_x | \psi_{m,\mathbf{k}} \rangle \langle \psi_{m,\mathbf{k}} | \hat{v}_y | \psi_{n,\mathbf{k}} \rangle$$

1. The Resonance (Denominator)

- $(E_m - E_n - \hbar\omega)$ dictates **when** a peak occurs
- Vertical interband transitions at specific k-points

The Kubo Formula

LINKING BAND STRUCTURE TO OPTICAL RESPONSE

$$\tilde{\sigma}_{xy}(\omega) \propto \frac{1}{\omega} \sum_{n,m} \int_{\text{BZ}} d\mathbf{k} \frac{f(E_{n,\{\mathbf{k}\}}) - f(E_{m,\mathbf{k}})}{E_{m,\mathbf{k}} - E_{n,\mathbf{k}} - \hbar\omega - i\Gamma} \times \langle \psi_{n,\mathbf{k}} | \hat{v}_x | \psi_{m,\mathbf{k}} \rangle \langle \psi_{m,\mathbf{k}} | \hat{v}_y | \psi_{n,\mathbf{k}} \rangle$$

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- $(E_m - E_n - \hbar\omega)$ dictates **when** a peak occurs
- Vertical interband transitions at specific $\mathbf{k}$ -points

2. The Topology (Matrix Elements)

- $\langle \hat{v}_x \rangle \langle \hat{v}_y \rangle$ relates to orbital angular momentum of the transition
- In altermagnets, this product survives the BZ sum

The Kubo Formula

LINKING BAND STRUCTURE TO OPTICAL RESPONSE

$$\tilde{\sigma}_{xy}(\omega) \propto \frac{1}{\omega} \sum_{n,m} \int_{\text{BZ}} d\mathbf{k} \frac{f(E_{n,\{\mathbf{k}\}}) - f(E_{m,\mathbf{k}})}{E_{m,\mathbf{k}} - E_{n,\mathbf{k}} - \hbar\omega - i\Gamma} \times \langle \psi_{n,\mathbf{k}} | \hat{v}_x | \psi_{m,\mathbf{k}} \rangle \langle \psi_{m,\mathbf{k}} | \hat{v}_y | \psi_{n,\mathbf{k}} \rangle$$

1. The Resonance (Denominator)

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- $\langle \hat{v}_x \rangle \langle \hat{v}_y \rangle$ relates to orbital angular momentum of the transition
- In altermagnets, this product survives the BZ sum

◆ Symmetry Note:

In standard AFMs,
 $\sigma_{xy}(\mathbf{k}) + \sigma_{xy}(\mathcal{PT}\mathbf{k}) = 0.$

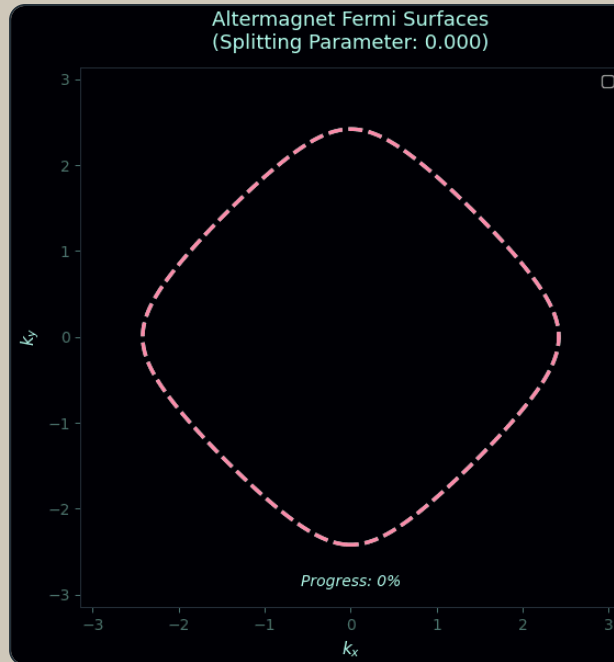
◆ Altermagnetism:

Alternating spin-polarization in $\mathbf{k}$ -space $\rightarrow$ net non-zero sum
for the off-diagonal elements.

The Kubo Formula

BAND SPLIT FERMİ SURFACE AND RESULTING KUBO INTEGRAL

$$\tilde{\sigma}_{xy}(\omega) \propto \dots \sum \int \frac{\Delta f}{\Delta E - \hbar\omega} \times \langle \hat{v}_x \rangle \langle \hat{v}_y \rangle$$



Standard AFM:

$$\sigma_{xy} = 0$$

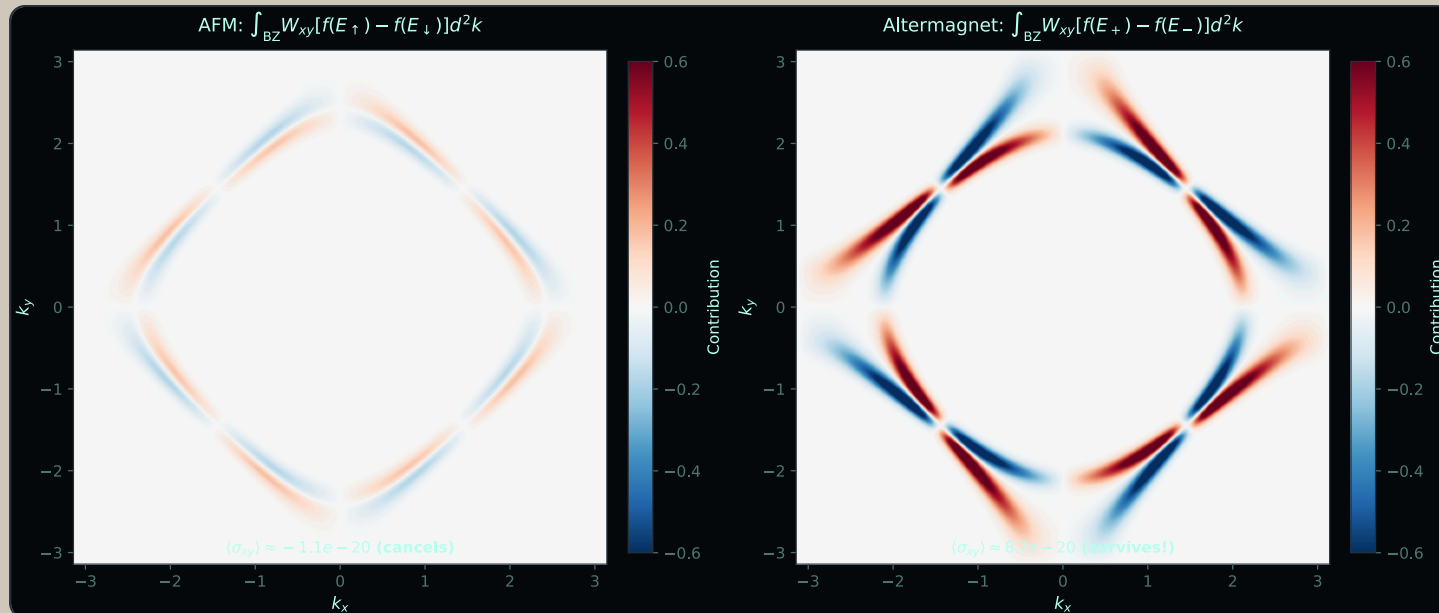
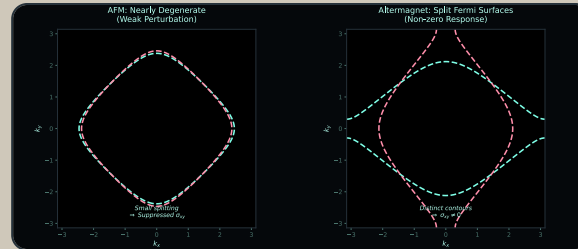


Altermagnet:

$$\sigma_{xy} \neq 0$$

The Kubo Formula

$$\tilde{\sigma}_{xy}(\omega) \propto \dots \sum \int \frac{\Delta f}{\Delta E - \hbar\omega} \times \langle \hat{v}_x \rangle \langle \hat{v}_y \rangle$$



Standard AFM:

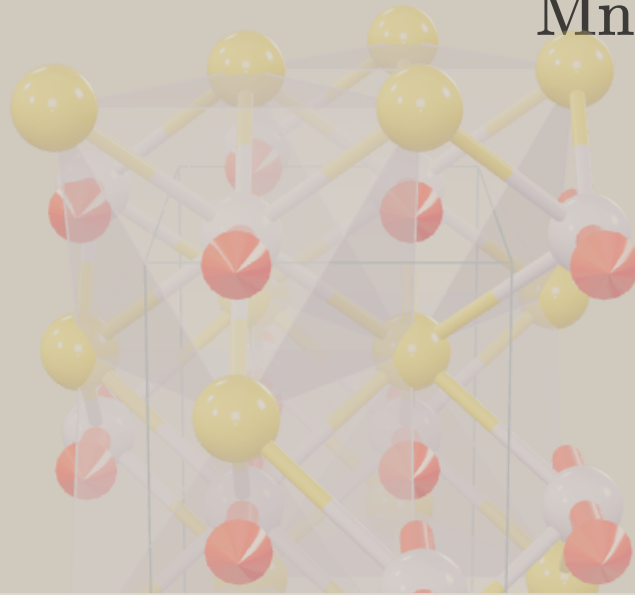
$$\sigma_{xy} = 0$$

Altermagnet:

$$\sigma_{xy} \neq 0$$

Off-Diagonal Tensor Elements

With the Examples of and
 MnTe CrSb



The Candidate: Hexagonal

SYMMETRY PROTECTION & ORDER PARAMETER α -MnTe

THE ORDER PARAMETER

Néel Vector :
 $\mathbf{L}$

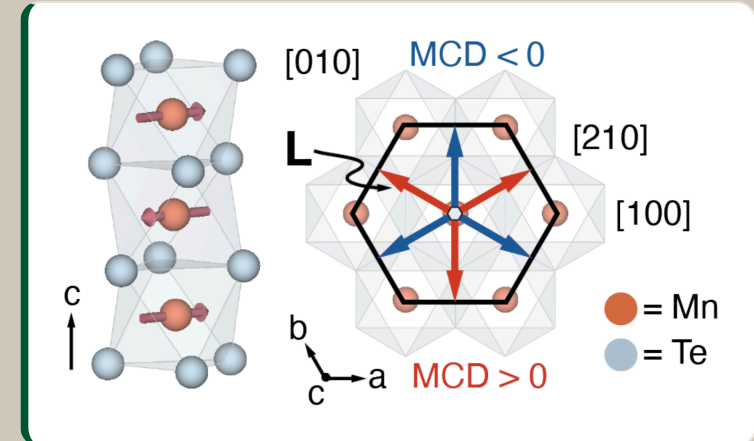
$$\mathbf{L} = \mathbf{M}_1 - \mathbf{M}_2$$

Probed via localized Mn²⁺ ions hosting high-spin states.

$$S = 5/2, L = 0$$

SYMMETRY & TOPOLOGY

- Two magnetic sublattices mapped onto each other via a **six-fold screw rotation**.
- Broken time-reversal symmetry () combined with spatial translation enforces a **d-wave altermagnetic state**.
- The crystal hosts a native **six-fold magnetocrystalline anisotropy (MCA)** pinning landscape.



Challenge for Production

To make altermagnets industrially viable for spintronics, we need large, macroscopically uniform domains. Knowing how external and internal stress fields alter this magnetic ordering dictates exactly how these devices must be fabricated and handled.

Coherent Néel Vector Steering

MAGNETO-OPTICAL MEASUREMENTS OF MCD, AND

ϕ_0 Δ

Measurements

- with the Neel vector orientation
- $\text{MCD} = \text{Im}[(r_{xy} - r_{yx})/2r_0] \propto L^3 \sin(3\theta_L)$
- the birefringence angle ϕ_0
- the birefringence amplitude $\Delta = \text{Re}[(r_{xx} - r_{yy})/2r_0]$
- Reference point: is the birefringence angle with respect to the $[100]$ direction and $r_0 = (r_{xx} + r_{yy})/2$

Connection to MOKE

The measurements are closely related to the optical conductivity tensor, and the dielectric tensor. The measurements relate through the intensities measured by MOKE:

$$I_\omega \propto MSD$$

$$I_{2\omega}(\phi) \propto \Delta \cdot \cos 2(\phi + \phi_0)$$

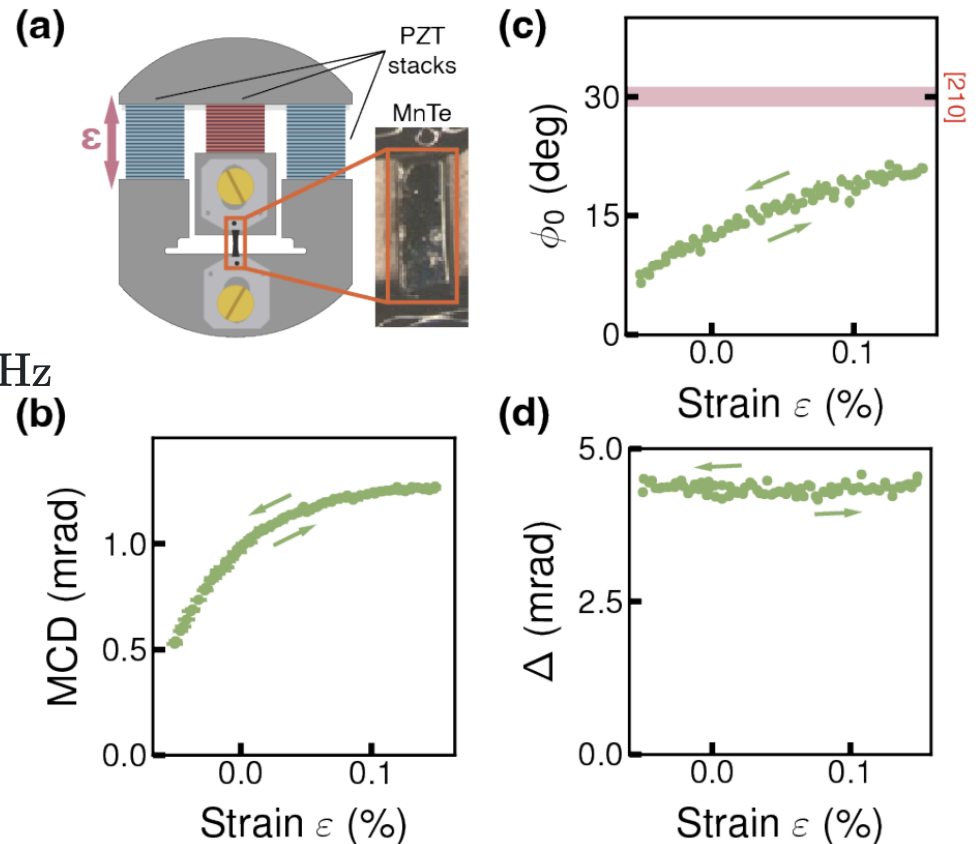
Coherent Néel Vector Steering

PROBING OFF-DIAGONAL ELEMENTS VIA IN-SITU STRAIN

Strain Dependence

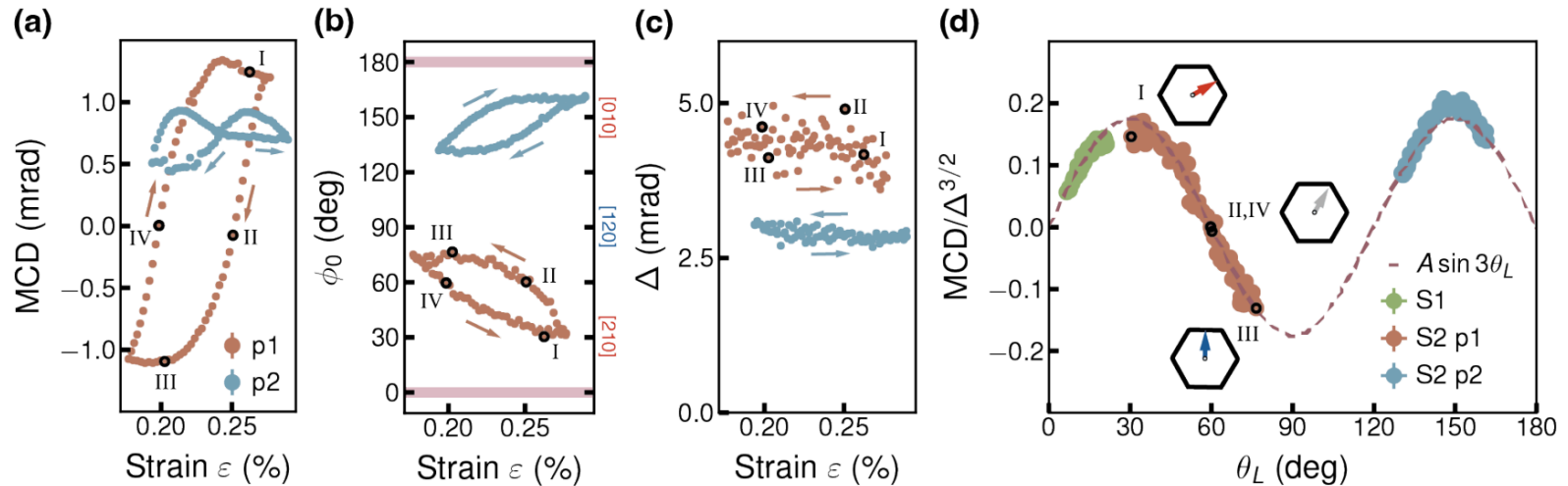
- and show hysteresis like behavior
- ϕ_0
Birefringence Amplitude is nearly constant over different strain Δ
- All measurements performed at

$$\omega/2\pi \approx 50 \text{ kHz}$$



Coherent Néel Vector Steering

PROBING OFF-DIAGONAL ELEMENTS VIA IN-SITU STRAIN



Strain Dependence

- and show hysteresis like behavior
- $MCD/\Delta^{3/2}$ is nearly constant over different strain

Néel Vector Orientation

- fits to $MCD \sin(\theta_L)$
- Néel Vector is the primary order parameter

Coherent Néel Vector Steering

PROBING OFF-DIAGONAL ELEMENTS VIA IN-SITU STRAIN

Magneto-Optical Trackers

By mapping the complex reflection coefficients, isolate elements linked directly to the off-diagonal dielectric tensor :

- $\epsilon_{xy}(\omega)$
- $\text{MCD} \propto L^3 \sin(3\theta_L)$
- ϕ_0
- $\Delta \propto L^2$

$$I_\omega \propto \text{MSD}$$

$$I_{2\omega}(\phi) \propto \Delta \cdot \cos[2(\phi + \phi_0)]$$

The Discovery

Under applied stress, **MCD** and ϕ_0 evolve continuously, while Δ remains perfectly constant.

Conclusion:

The stress is not changing the size of the alternating magnetic order parameter (I_ω). It is continuously rotating its direction (θ_L).

This behavior matches a perfect energetic competition model between the native crystal 6-fold Magnetocrystalline Anisotropy and the applied 2-fold strain anisotropy.

Coherent Néel Vector Steering

PROBING OFF-DIAGONAL ELEMENTS VIA IN-SITU STRAIN

Magneto-Optical Trackers

By mapping the complex reflection coefficients, isolate elements linked directly to the off-diagonal dielectric tensor $\epsilon_{xy}(\omega)$:

- $\text{MCD} \propto L^3 \sin(3\theta_L)$
- ϕ_0 : Birefringence angle
- Δ : Birefringence amplitude $\propto L^2$

DON'T STRESS YOUR ALTERMAGNET!

$$I_{2\omega}(\phi) \propto \Delta \cdot \cos[2(\phi + \phi_0)]$$

The Discovery

Under applied stress, **MCD** and ϕ_0 **evolve continuously**, while Δ **remains perfectly constant**.

Conclusion:

The stress is not changing the size of the altermagnetic order parameter (L). It is

between the native crystal 6-fold Magnetocrystalline Anisotropy and the applied 2-fold strain anisotropy.

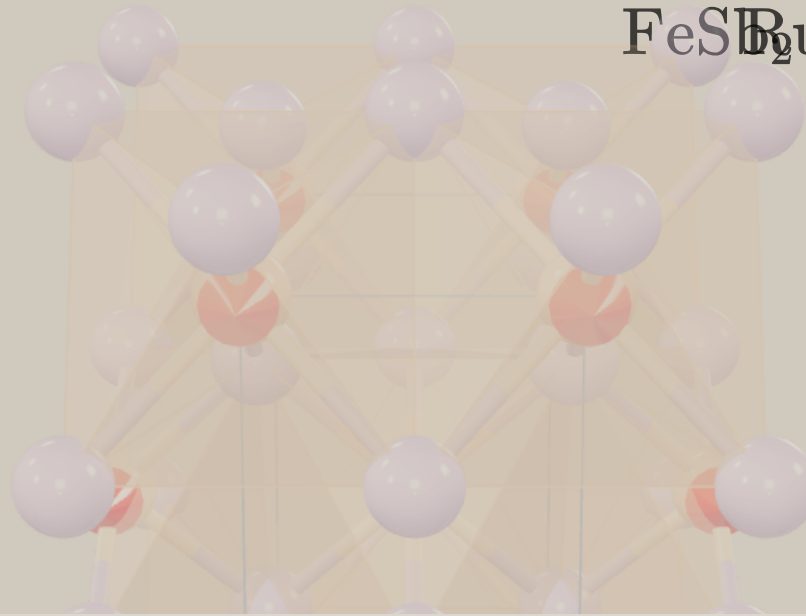
The Candidate:

DUMMY TEXT

CrSb₂

Diagonal Tensor Elements

With the Examples of



From Narrow-Gap Semiconductor to Metallic Altermagnet: Optical Fingerprints of Co-Doped FeSb₂

R. Mathew Roy^{1,*}, M. Povolotskiy¹, J. Kirschke², C. Prange¹, Y. Xia³, V. Sundaramurthy⁴, P. Puphal⁴,
M. Pinterić^{1,5}, M. van de Loo², A. Kreyssig², T. Zhang³, A. E. Böhmer², M. Dressel¹, and M. Wenzel^{1,†}

¹*Physikalisches Institut, Universität Stuttgart, Pfaffenwaldring 57, 70569 Stuttgart, Germany*

²*Experimental Physics IV, Department of Physics and Astronomy,
Ruhr University Bochum, 44801 Bochum, Germany*

³*Institute of Theoretical Physics, Chinese Academy of Sciences, Beijing 100190, China*

⁴*Max Planck Institute for Solid State Research, Heisenbergstraße 1, 70569 Stuttgart, Germany*

⁵*Faculty of Civil Engineering, Transportation Engineering and Architecture,
University of Maribor, SI-2000 Maribor, Slovenia*

(Dated: May 7, 2026)

The realization of bulk metallic altermagnetism has remained elusive despite the growing number of candidate materials. Here, we present evidence that moderate cobalt substitution ($\sim 15\%$) drives the correlated narrow-gap semiconductor FeSb₂ into a metallic altermagnetic state persisting up to room temperature. The infrared optical conductivity reveals low-energy interband transitions near 0.1 eV that emerge upon doping and grow with Co concentration. Density functional theory calculations show that these transitions originate exclusively from altermagnetic spin ordering, with spin split bands (~ 0.2 eV) of non-relativistic origin, together with spin-orbit coupling induced band splitting of the order of ~ 5 meV near the Fermi level. Co substitution further leads to Fano lineshapes and mode mixing in the infrared-active phonons, reflecting enhanced electron-phonon coupling and local inversion symmetry breaking, while leaving the altermagnetic spin symmetry intact. Our results establish carrier-tuned FeSb₂ as a platform for exploring metallic *d*-wave altermagnetism and its coupling to lattice degrees of freedom.

The Parent Compound

AS A NATURAL ALTERMAGNET CANDIDATE



Orthorhombic structure

Pnnm

- **(Primitive):** Primitive, no additional lattice points except at corners
- **(1st axis):** n-glide plane perpendicular to the a-axis. Reflection followed by translation: $t = \frac{b+c}{2}$
- **(2nd axis):** n-glide plane perpendicular to the b-axis. Reflection followed by translation: $t = \frac{a+c}{2}$
- **(3rd axis):** A standard Mirror plane perpendicular to the c-axis. This is a simple reflection without any translation.

Sublattice is invariant under two-fold rotation + translation

Fe

Makes a **natural altermagnet candidate**

$$[C_2 \parallel C_{2x}t]$$

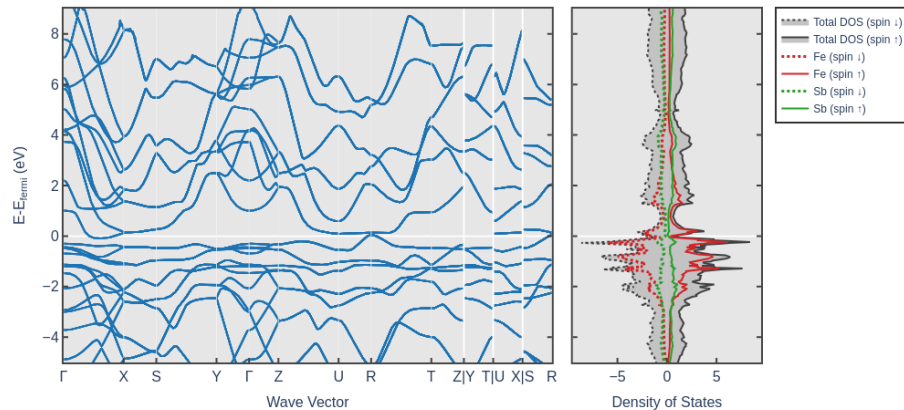
The necessary Symmetry Condition

FOR THE EXAMPLE OF THE CRYSTAL UNIT CELL
 FeSb_2

The Parent Compound

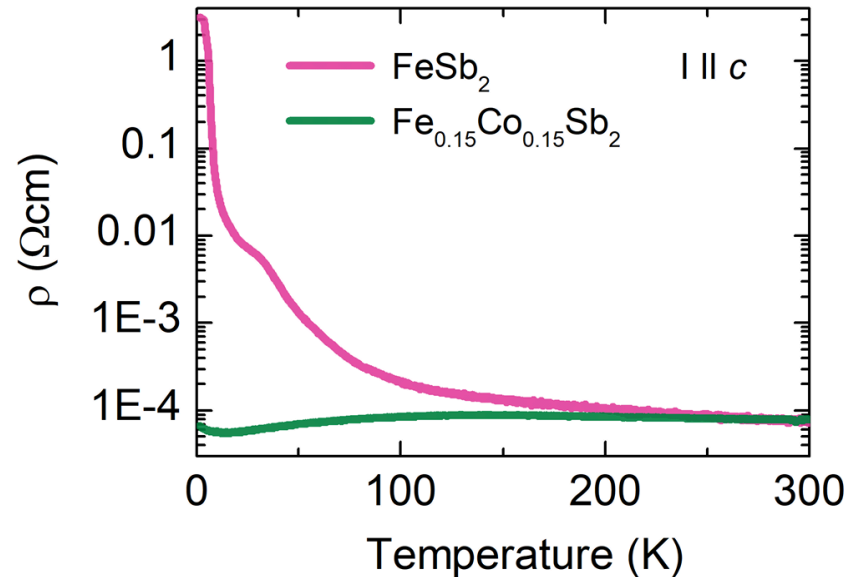
CORRELATED NARROW-GAP SEMICONDUCTOR

BAND STRUCTURE



- Transport gap of approximately 30 meV
- Optical gap of roughly 120 meV

RESISTIVITY

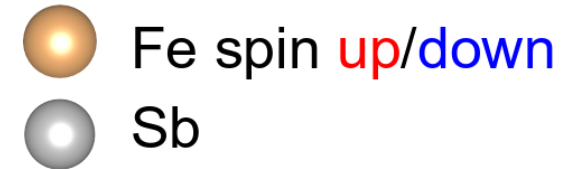


- No long range magnetic order
- Rising resistivity at low temperatures

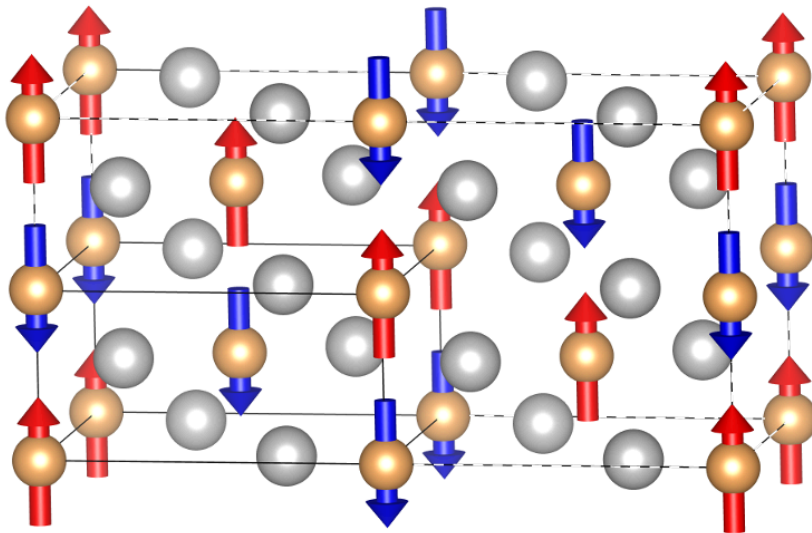
The Parent Compound

CORRELATED NARROW-GAP SEMICONDUCTOR

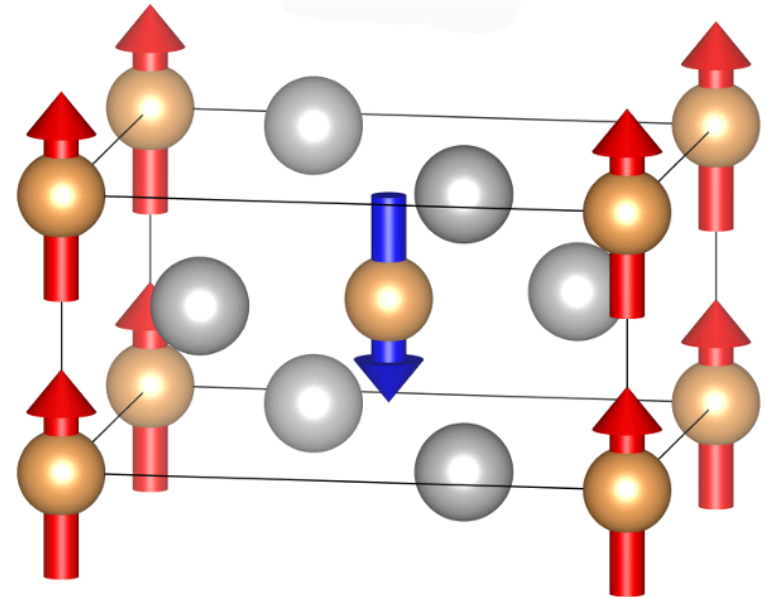
- Transport gap of approximately 30 meV
- Optical gap of roughly 120 meV
- No long range magnetic order
- Rising resistivity at low temperatures



ANTIFERROMAGNETIC (AFME) ORDER



ALTERNATING (AFMO) ORDER



Cobalt-doped

STABILATION VIA FeSb_2 DOPING WITH ELECTRONS

Adding electrons/holes

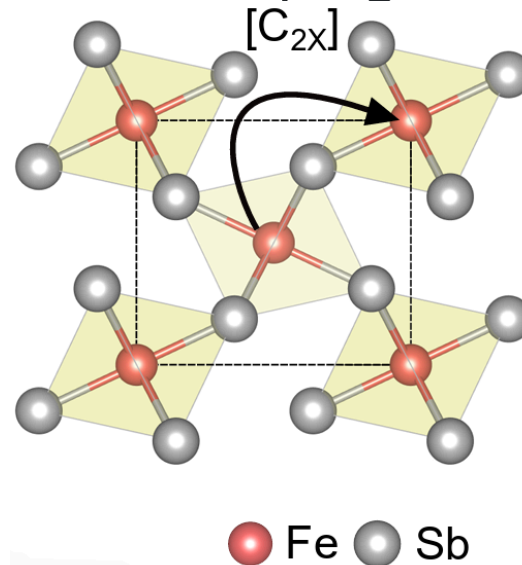
- Adds impurities
- Quenches the spin fluctuations
- Stabilize the AFMo order
- Adding holes has similar effect (doping Cr)

Result

- Crystal structure is preserved
- Semiconducting gap is closed
- Persisting up to room temperature

 Schematic showing the effect of electron doping

 Schematic showing the effect of hole doping

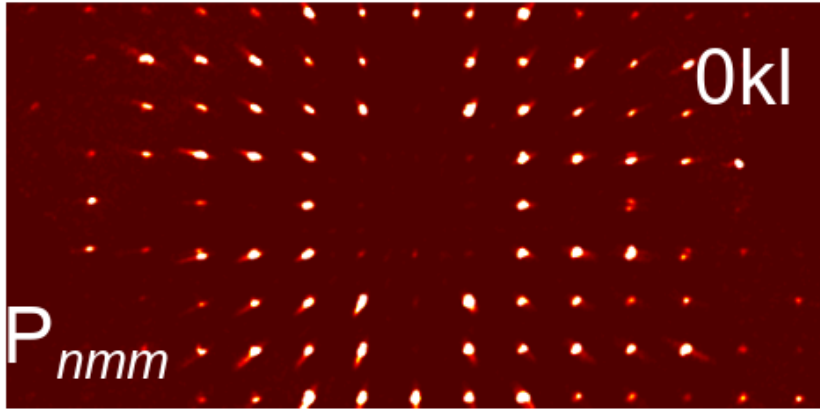


Cobalt-doped FeSb₂

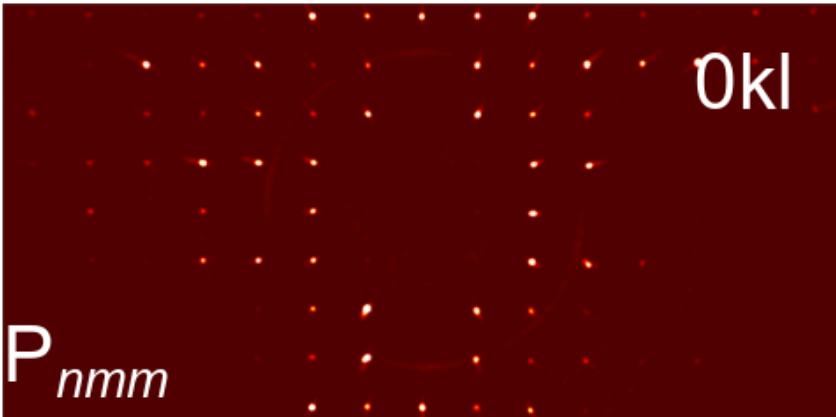
CLOSING THE GAP AND STABILIZING THE ALTERMAGNETIC STATE

X-RAY DIFFRACTION PATTERNS

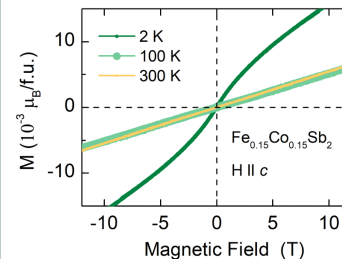
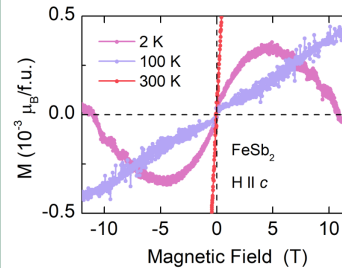
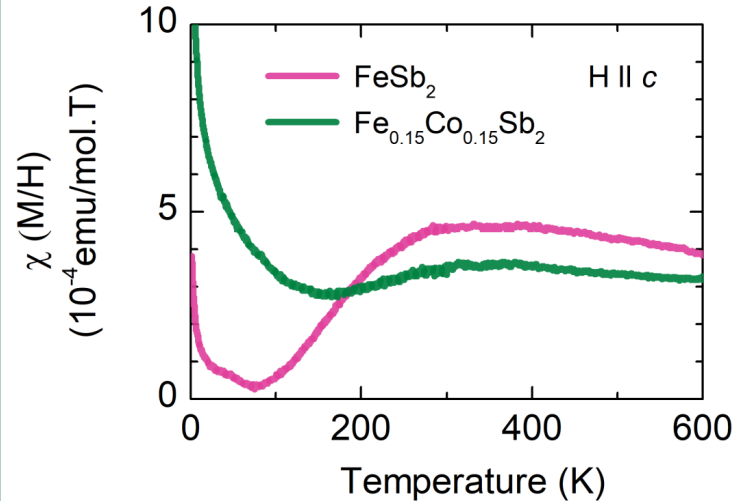
FeSb₂



Fe_{0.85}Co_{0.15}Sb₂



CRYSTAL STRUCTURE



Effects of Cobalt Doping

TRANSFORMATION FROM CORRELATED SEMICONDUCTOR TO ANISOTROPIC METAL

The Semiconductor-to-Metal Transition

- is a narrow-gap semiconductor (~ 0.3 eV) with high resistivity at low temperatures.
- With **Cobalt substitution** (FeSb_2), you inject extra electrons into the system. This extra charge **collapses the semiconducting gap**. $\text{Fe}_{0.85}\text{Co}_{0.15}\text{Sb}_2$
- In the optical conductivity, this is proven by the emergence of a **Drude peak** along with new mid-infrared interband transitions $\sigma_1(\omega)$ at 0.1 eV.

Drude peak: a sharp spike in absorption centered at zero energy, which is the universal optical signature of free, mobile conduction electrons

Persistent Structural and Optical Anisotropy

Quenching Spin Fluctuations

Effects of Cobalt Doping

TRANSFORMATION FROM CORRELATED SEMICONDUCTOR TO ANISOTROPIC METAL

The Semiconductor-to-Metal Transition

Persistent Structural and Optical Anisotropy

- Metallic transition does not destroy directional preferences. The optical response is still highly dependent on whether the light is polarized along the crystal's a -axis or c -axis.
- Unscreend plasma frequency (ω_p) along the a -axis (2.23 eV) higher than c -axis (1.95 eV). Since ω_p^2 is proportional to the carrier density (n) divided by the effective mass (m^*), this means the electrons are physically "lighter" or more densely packed along the a -axis.
- Crucial, because altermagnets natively rely on anisotropic transport properties.

Quenching Spin Fluctuations

Effects of Cobalt Doping

TRANSFORMATION FROM CORRELATED SEMICONDUCTOR TO ANISOTROPIC METAL

The Semiconductor-to-Metal Transition

Persistent Structural and Optical Anisotropy

Quenching Spin Fluctuations

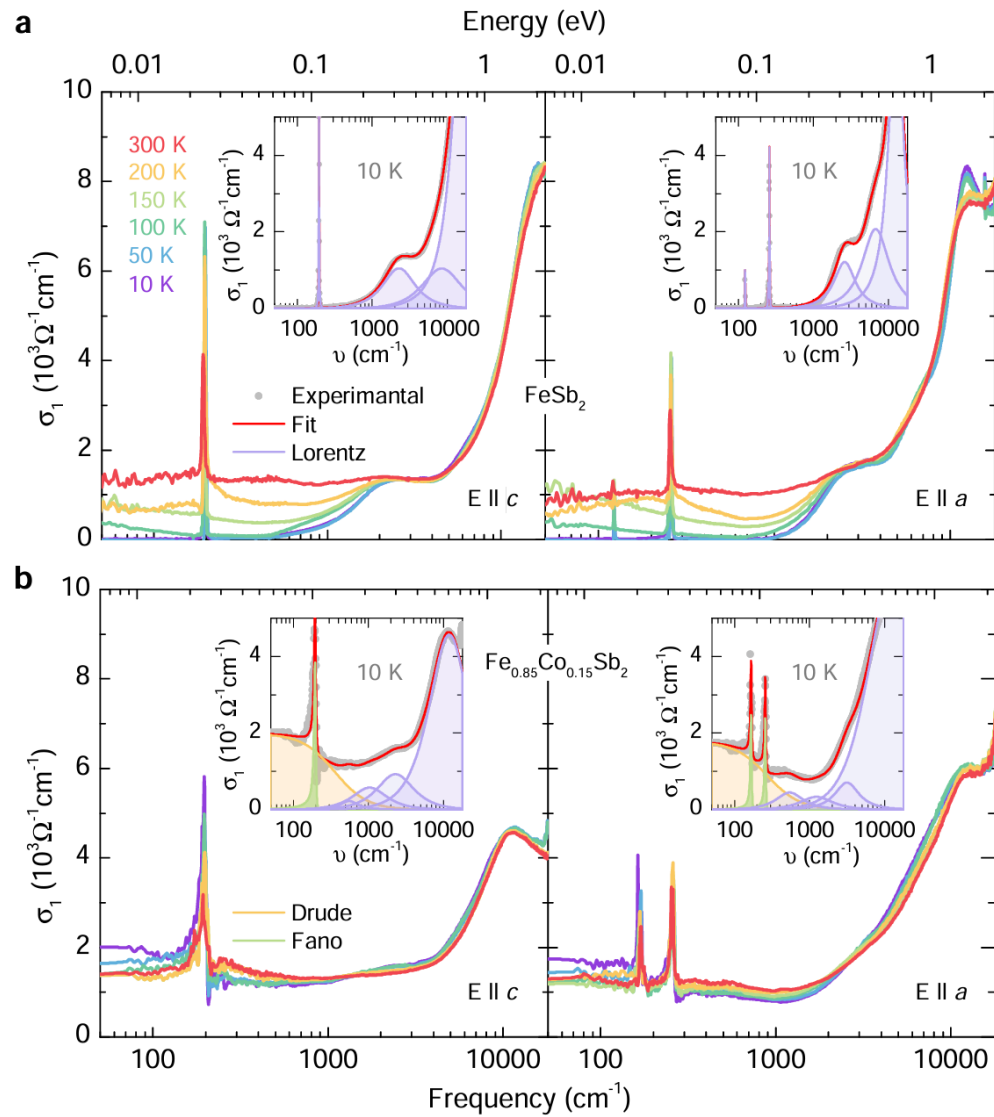
- Adding impurities/dopants to a pristine crystal creates atomic disorder, increasing the electron scattering rate ($1/\tau$), making the material dirtier and more resistive.
- But: scattering rate actually drops drastically to less than half of the parent compound's value (e.g., along the a -axis at 300 K, from 1221 cm^{-1} down to 336 cm^{-1}).
- Therefore, the dominant source of electrical resistance was highly violent, incoherent spin fluctuations.

Cobalt doping introduces a stable magnetic environment that suppresses these fluctuations so the electrons can flow much more smoothly through the lattice, despite the Cobalt impurities.

Effects of Cobalt Doping

TRANSFORMATION FROM CORRELATED SEMICONDUCTOR TO
ANISOTROPIC METAL

text



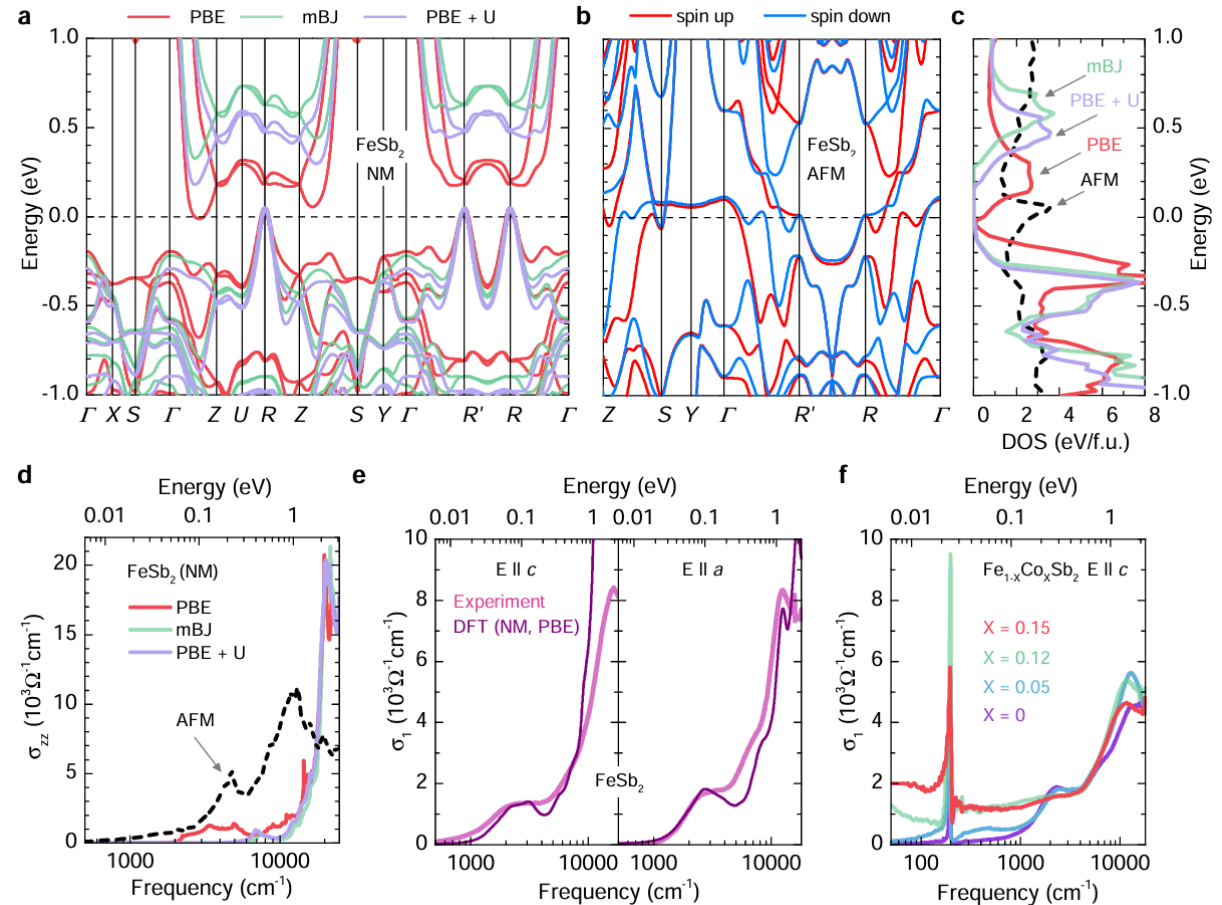
Preparing the DFT Simulation

SELECTION OF EXCHANGE-CORRELATION FUNCTIONAL AND MAGNETIC ORDER

- No matching at DOS_{AF}
- No matching Band structure
- No matching optical conductivity

Conclusion:

The AFM state is not the ground state of FeSb_2



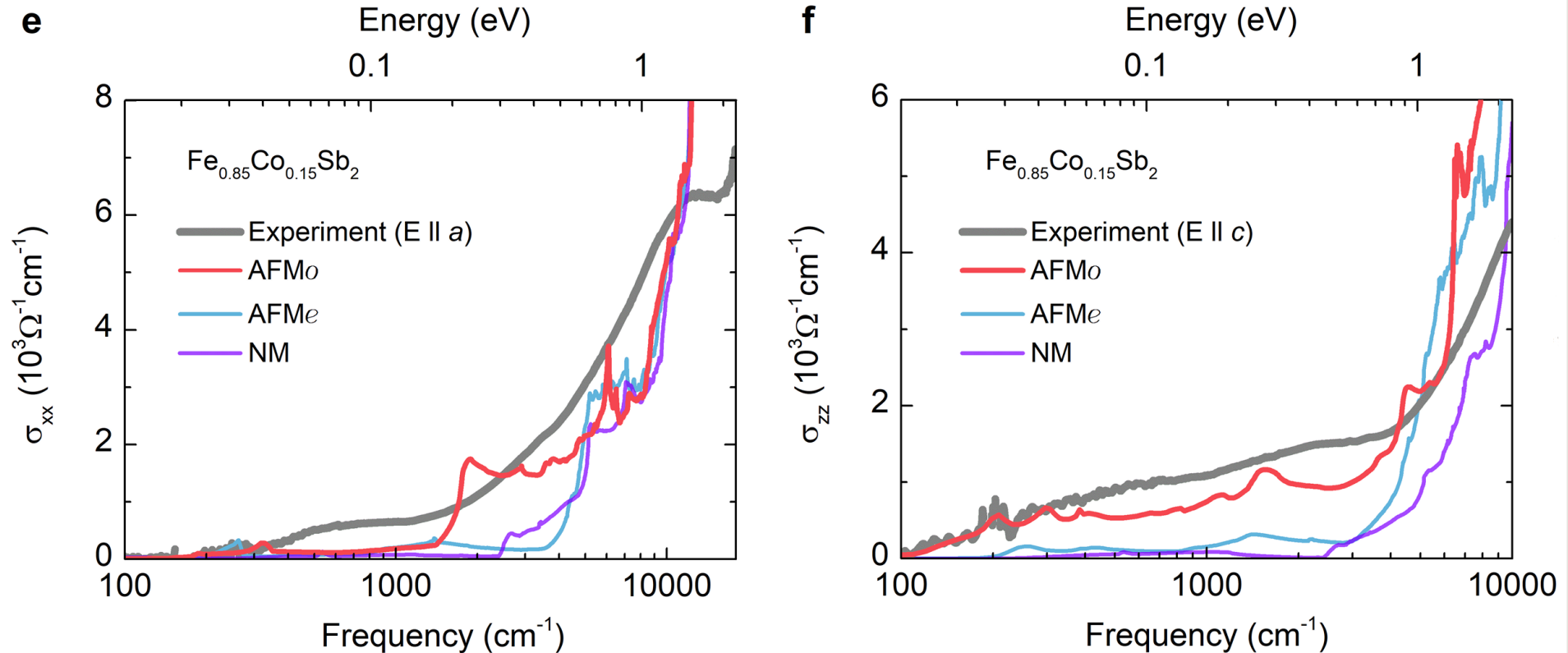
PBE simulates non-magnetic best .

The underestimation of band gaps aligns with itinerant spin fluctuations.

FeSb₂

Identifying the Altermagnetic State

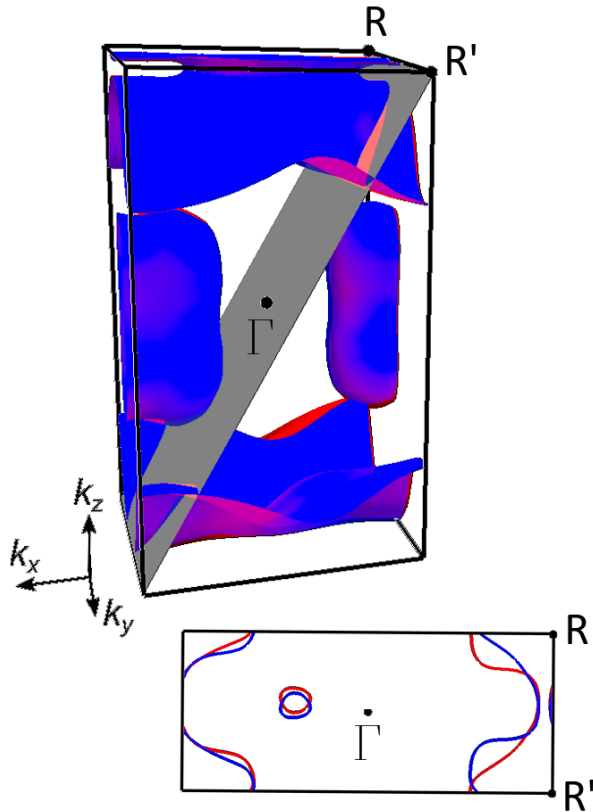
THE DFT COMPARISON (NM VS. AFME VS. AFMO)



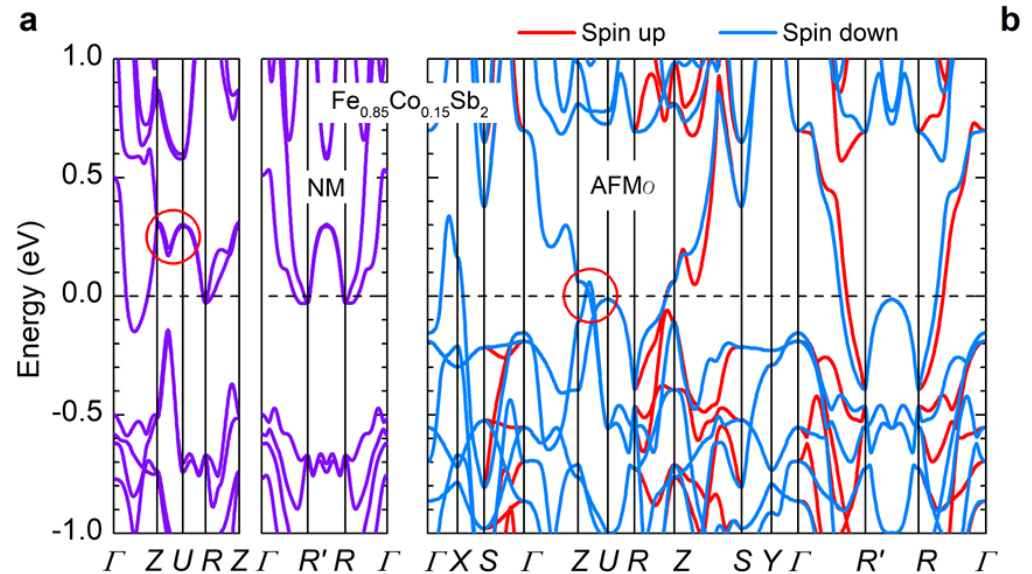
Only the **AFM₀** (red line) state shows a good agreement with the experimental data.

Identifying the Altermagnetic State

NON-RELATIVISTIC SPIN SPLITTING



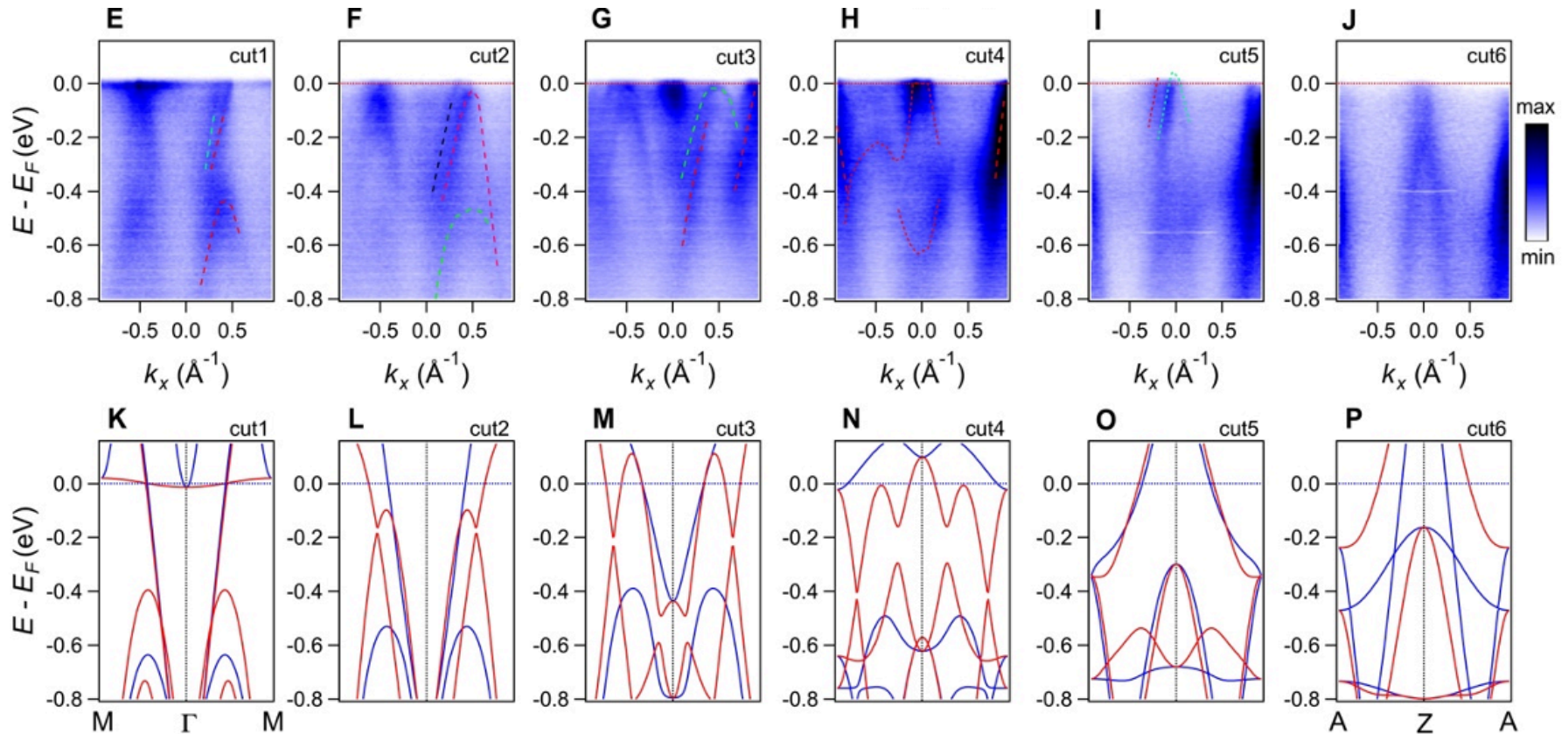
- Clear spin-splitting of the Fermi surface
- Alternating spin-polarization across the Brillouin zone



Controversy: A Model Altermagnet?

RuO_2 EARLY PREDICTION OF ALTERMAGNETISM

Surface Probe (ARPES): Evidence for Spin-Splitting



Controversy: A Model Fermi-Liquid?

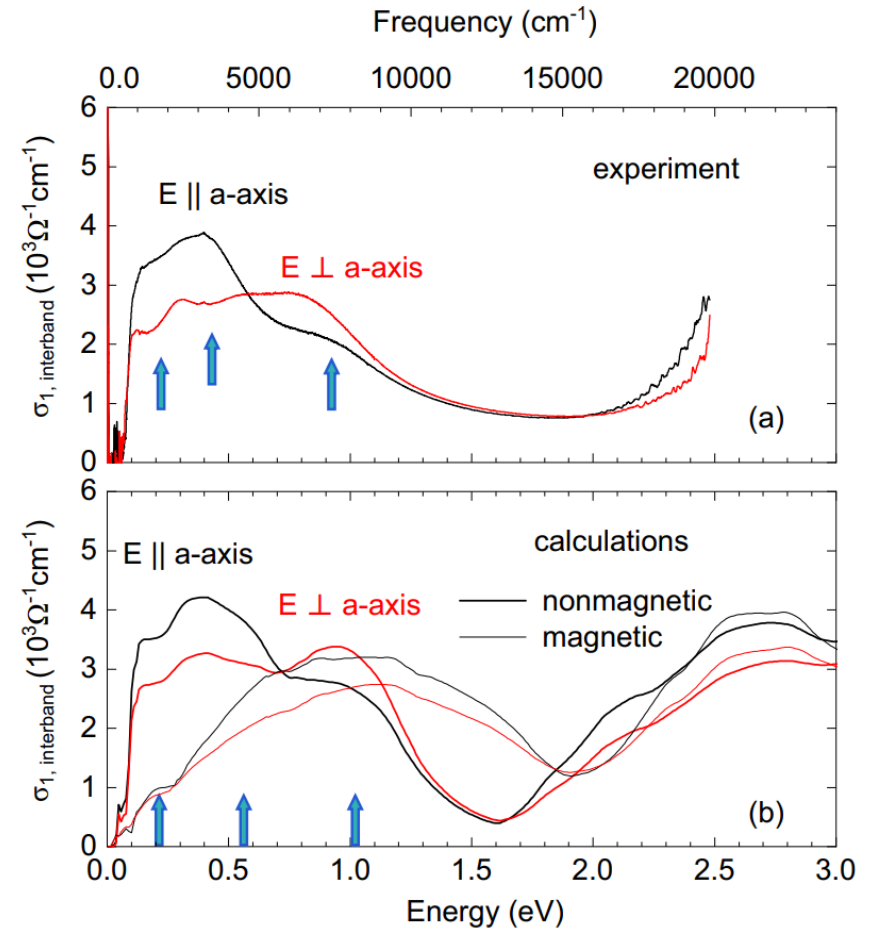
RuO_2 FERMILIQUID BEHAVIOR IN BULK



Experiment matches NON-magnetic model

test

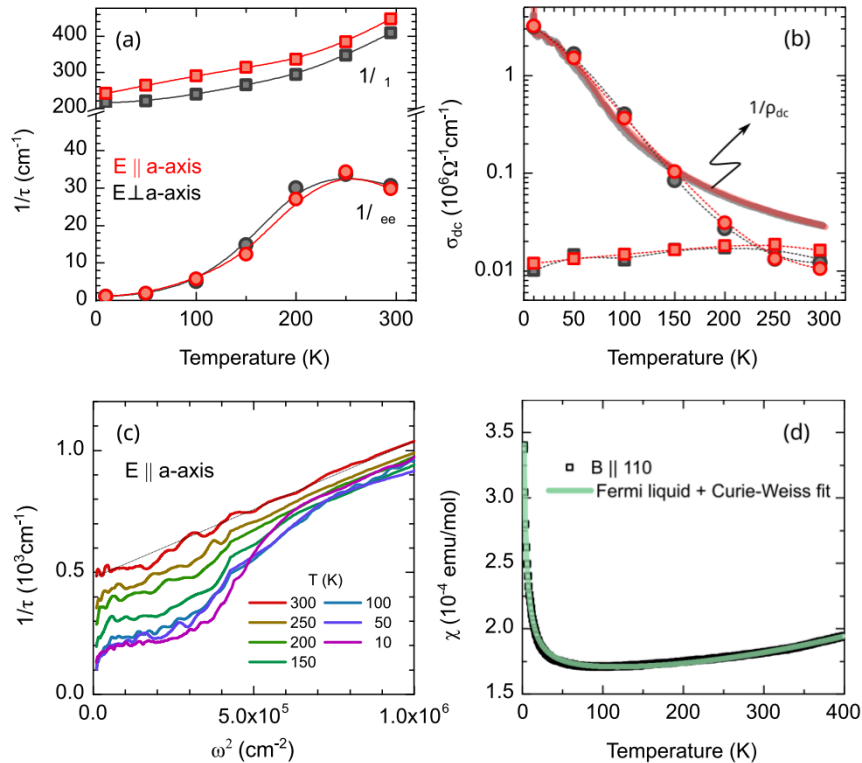
- Sharp **Pauli Edge** around 0.1 eV
- Minimum around 1.5 – 2.0 eV
- Only **non-magnetic** DFT calculation matches



Controversy: A Model Fermi-Liquid!

RuO_2 A PARAMAGNETIC FERM LIQUID

Other characteristics match fermi liquids too!



Magnetic Susceptibility Data

Figure (d):

- Excellent fit to a model combining Pauli paramagnetism (T^2 dependence)
- Small impurity tail

Phase Angle

Figure (c)/(e):

- Quadratic dependence (ω^2)
- hallmark of a Fermi liquid

Controversy: A Model Fermi-Liquid!

RuO2 PARAMAGNETIC FERM LIQUID

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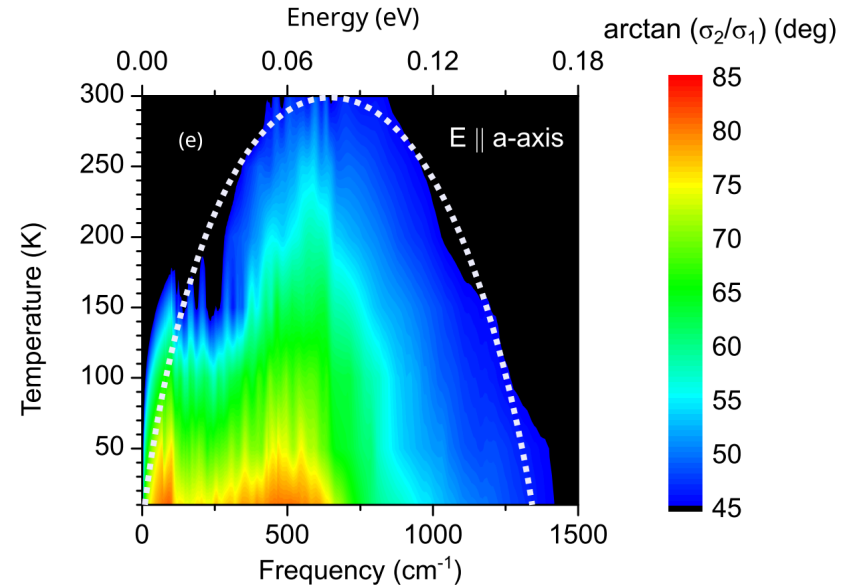
Figure (d):

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Figure (c)/(e):

- Quadratic dependence (ω^2)
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Summary

MOKE

- Single
- High Sensitivity ^{ω}

MOKKA

- Broad Band
- Low Sensitivity

Mueller

- Model dependent
- Regression Task

Summary

MOKE

MOKKA

Mueller

MnTe

- info

CrSb

- info

FeSb₂

- info

RuO₂

- info